THE UNIVERSITY OF SHEFFIELD

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Convex and Integer Optimization for Microgrid Energy Scheduling with Battery Storage

Electrical and Electronic Engineering

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Abstract

This dissertation investigates microgrid energy scheduling with battery storage, focusing on the role of discrete decision-making in optimisation models. A grid-connected microgrid is considered, comprising a dispatchable generator, battery energy storage, and fixed demand. The interaction with the main grid is modelled flexibly, allowing both electricity import and export depending on economic parameters, thereby enabling price-driven energy arbitrage.

Three optimisation formulations are developed and compared: a convex baseline model, a regularised convex model incorporating sparsity and switching penalties, and a mixed-integer model with explicit binary commitment and start-up decisions. A unified and fair comparison framework is established to ensure consistent evaluation across all models in terms of cost, operational behaviour, and computational performance.

The study aims to determine under what conditions mixed-integer formulations are necessary and to what extent regularized convex models can approximate discrete operational behavior. Numerical experiments are conducted using both synthetic 168-hour benchmark data and real-world UK National Grid electricity price data. The results show that regularized convex models can reproduce key behavioral characteristics of mixed-integer solutions, such as reduced generator utilization and limited switching, within certain parameter regimes. However, they are unable to capture hard logical constraints, including minimum generation thresholds and discrete start-up events.

The findings highlight a fundamental trade-off between modelling fidelity and computational efficiency. Mixed-integer formulations provide exact representations of operational logic at the cost of increased complexity, whereas regularised convex models offer scalable approximations with reduced computational burden. These insights provide guidance on when discrete optimisation is essential in microgrid scheduling and when continuous relaxations are sufficient.

Individual Contribution

This dissertation presents work undertaken independently by the author as part of the Individual Project module in the School of Electrical and Electronic Engineering at the University of Sheffield.

The problem formulation, modelling approach, implementation, numerical experiments and analysis were developed and conducted by the author. Three optimisation formulations were designed and implemented: a convex baseline model, a regularised convex model incorporating sparsity and switching penalties, and a mixed-integer formulation including binary commitment and start-up variables. The selection of objective functions, constraints and comparison metrics was carried out by the author.

The optimisation models were implemented using AMPL and solved using a commercial optimisation solver. Post-processing, visualisation and comparative analysis were performed using MATLAB. The synthetic benchmark data used for controlled experiments were generated by the author. Real-world electricity price data were obtained from publicly available UK National Grid sources, and were preprocessed, formatted and integrated into the modelling framework by the author.

The comparative evaluation framework, including the fair comparison protocol and performance metrics such as total cost, generator usage, switching behaviour and computational time, was designed and executed by the author. All numerical results, figures and tables presented in this dissertation were produced by the author.

No external code bases were used in the development of the optimisation models. Standard optimisation software and publicly available datasets were used where appropriate and are cited accordingly.

Use of Generative AI Generative AI tools were used to support language refinement, structural organisation, LaTeX formatting, technical explanations, and code calibration. They were not used to generate original results or make final technical decisions. All modelling, coding, simulations, data processing, figures, result interpretation, and final academic decisions were carried out, checked, and approved by the author.

Acknowledgements

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I would also like to thank my second marker, Dr. Yuanbo Nie, for his constructive advice during the interim report meeting. His suggestions helped me to refine the scope of the project and to shift the main focus of the second semester towards comparing the differences among the three optimisation models, rather than attempting to build a truly large-scale model.

Finally, I would like to thank the School of Electrical and Electronic Engineering at the University of Sheffield for providing a supportive academic environment and the resources necessary for completing this project.

Contents

1 Introduction

1.1 Background

The increasing penetration of renewable energy sources and distributed energy resources has accelerated the development of microgrid systems, which coordinate local generation, storage, and demand to improve reliability and efficiency [1], [2].

Battery energy storage systems (BESS) play an important role by shifting energy across time, charging during low-price periods and discharging during high-price periods [3], [4].

Optimisation-based energy management is widely used in microgrids. Convex optimisation offers strong computational efficiency, while mixed-integer programming is typically used to represent discrete decisions such as generator commitment, start-up events, and battery operating modes [5], [6]. Because mixed-integer formulations are computationally more demanding, recent work has explored convex approximations and regularisation-based methods as intermediate approaches [7], [8].

This project considers a grid-connected microgrid with a dispatchable generator, battery storage, and fixed demand, and compares convex, regularised convex, and mixed-integer optimisation approaches within a unified framework.

1.2 Objectives

The primary aim of this project is to investigate the role of discrete decision-making in microgrid energy scheduling and to assess when mixed-integer optimisation formulations are necessary. To achieve this aim, the objectives are:

- To develop convex, regularised convex, and mixed-integer optimisation models for a grid-connected microgrid including grid interaction, generator operation, and battery dynamics.
- To establish a fair comparison framework with common system conditions, input data, and evaluation metrics.
- To conduct numerical experiments using synthetic benchmark data and UK electricity price data.
- To compare the models in terms of cost, operational behaviour, and computational performance, and to determine when regularised formulations provide a sufficient approximation to mixed-integer solutions.

2 Theory and Literature Review

2.1 Convex Optimisation

Convex optimisation provides the continuous baseline model used in this dissertation. A problem is convex when both the objective function and feasible set are convex, so any local optimum is also globally optimal [9]. A standard convex optimisation problem is

$$\min_{x} f_0(x) \quad \text{s.t.} \quad f_i(x) \leq 0,\ Ax = b, \tag{2.1}$$

where $f_0(x)$ and $f_i(x)$ are convex functions and the equality constraints are affine.

In microgrid energy scheduling, convex models arise when decision variables such as grid import, generator output, battery power and state of charge are continuous. With linear power balance and battery dynamics, the problem is typically a linear or quadratic programme that can be solved efficiently.

Convex optimisation is widely used in microgrid energy management and storage dispatch because of its computational efficiency and global optimality guarantees [10], [11]. However, it cannot explicitly model discrete decisions such as generator on/off states, start-up events or strict charge/discharge exclusivity. Therefore, it is used here as an efficient baseline, motivating the regularised and mixed-integer models introduced later.

2.2 Mixed-Integer Optimisation

To address the limitations of convex formulations in representing discrete operational decisions, mixed-integer optimisation (MIO) is widely used in power systems and microgrid scheduling. Unlike purely convex models, MIO introduces binary variables to represent logical decisions such as on/off states, start-up events and mutually exclusive operating modes.

A general mixed-integer optimisation problem can be written as

$$\min_{x,z} \ f(x,z) \quad \text{s.t.} \quad g(x,z) \leq 0,\ z \in \{0,1\}, \tag{2.2}$$

where x denotes continuous variables and z denotes binary decision variables. The integer variables make the problem non-convex and generally more difficult to solve.

In microgrid energy management, mixed-integer formulations are commonly used to capture generator commitment, start-up decisions and discrete battery operating modes. Recent reviews show that mixed-integer and related optimisation methods remain central in microgrid scheduling [12], [13]. MILP-based formulations have also been applied to practical problems such as battery degradation-aware scheduling and distributed energy coordination [14], [15].

However, this modelling fidelity increases computational complexity. Mixed-integer problems are harder to solve than continuous convex problems because the solver must search over combinations of discrete decisions as well as continuous variables. Many MIP problems are NP-hard, so solution time can grow rapidly with the number of binary variables and the scheduling horizon.

Therefore, although mixed-integer optimisation can accurately represent discrete operational behaviour, its computational burden motivates the study of regularised convex formulations in the next section.

2.3 Regularisation Techniques

While mixed-integer optimisation can explicitly model discrete operational decisions, its computational complexity motivates continuous approximation methods. Regularisation is one such approach, where penalty terms are added to the original objective to encourage structured behaviour without introducing binary variables.

A regularised optimisation problem can be written as

$$\min_{x} \; f(x) + \lambda R(x), \tag{2.3}$$

where $f(x)$ is the original objective, $R(x)$ is the regularisation term, and λ controls the penalty strength.

The regularisation term discourages undesirable solution patterns. For example, an ℓ_1 penalty promotes sparsity by pushing some decision variables towards zero, which can approximate on/off behaviour without binary variables. In microgrid scheduling, ℓ_1 penalties can reduce generator usage, while total-variation penalties can discourage rapid changes between time steps. These provide tractable convex approximations to behaviours that would otherwise require combinatorial modelling [12], [13], [16].

However, regularisation remains a soft approximation. It can encourage discrete-like behaviour, but cannot strictly enforce logical constraints such as exact start-up events, minimum operating times, or mutually exclusive operating modes. This trade-off between computational efficiency and modelling fidelity motivates the comparative framework used in this dissertation.

2.4 Microgrid Scheduling Literature

Microgrid energy scheduling has been widely studied for systems with renewable generation, battery storage and flexible demand. Its main objective is usually to minimise operating cost while satisfying demand and operational constraints over a finite time horizon [12], [17], [18].

Existing studies commonly adopt convex or mixed-integer optimisation. Convex models are suitable for continuous decisions such as grid import, generator output, battery power and state of charge, while mixed-integer models explicitly represent discrete logic such as generator commitment, start-up events, battery operating modes and demand response [12], [19], [20].

Recent work has further improved realism and tractability through battery degradation modelling, uncertainty-aware scheduling, model predictive control and rolling-horizon methods [16], [17], [21], [22], [23], [24].

Relaxation and regularisation-based methods offer an intermediate approach between continuous convex optimisation and exact mixed-integer modelling. They can approximate some binary-variable effects, such as sparse operation or reduced switching, without explicitly introducing integer variables [25], [26]. However, they provide soft approximations rather than exact logical guarantees.

Several microgrid studies have applied mixed-integer programming to unit commitment, battery operation and demand response [27], [28]. Overall, the literature shows a trade-off between modelling fidelity and computational efficiency, but direct comparisons between convex, regularised convex and mixed-integer formulations under the same microgrid setting remain limited. This motivates the comparative framework adopted in this dissertation.

2.5 Research Gap

Existing studies on microgrid energy scheduling highlight a trade-off between modelling fidelity and computational efficiency. Convex optimisation is efficient and scalable, but it cannot explicitly represent discrete operational decisions such as generator commitment, start-up events, or battery operating modes. Mixed-integer formulations can model these decisions directly, but at the cost of higher computational complexity [12], [16], [17].

Recent studies have explored relaxation and regularization-based methods as tractable alternatives to exact mixed-integer modeling, but their ability to reproduce mixed-integer scheduling behavior remains insufficiently quantified [25], [26].

Therefore, the main research gap is the lack of a fair and systematic comparison between convex, regularized convex, and mixed-integer formulations under the same microgrid scheduling framework. This dissertation addresses this by comparing their cost, computational performance and ability to represent discrete operational behaviour.

3 Methodology

3.1 System Model

This dissertation considers a grid-connected microgrid with a dispatchable generator, a battery energy storage system (BESS), and a known time-varying demand profile. The grid is assumed to be always available, with electricity imported at time-varying prices. Local flexibility is provided by the generator and battery: the generator operates when economically beneficial, while the battery charges during low-price periods and discharges during high-price periods. The full system schematic and parameter definitions are provided in Appendix A.

The scheduling horizon is divided into N discrete time intervals, with

$$T = \{1, \dots, N\}.$$

At each time period $t \in T$, the electrical demand D_t must be supplied by grid import, generator output, and battery discharge. The battery may also absorb energy through charging. The power balance is therefore

$$P_t^{grid} + P_t^g + P_t^d = D_t + P_t^c, \quad \forall t \in T, \tag{3.1}$$

where P_t^{grid} is the grid import power, P_t^g is the generator output, and P_t^c and P_t^d denote the battery charging and discharging power, respectively.

The battery dynamics are described by the state of charge (SoC), which evolves according to

$$E_t = E_{t-1} + \eta_c P_t^c \Delta t - \frac{1}{\eta_d} P_t^d \Delta t, \tag{3.2}$$

subject to operational limits

$$E_{\min} \leq E_t \leq E_{\max}, \quad \forall t \in T, \tag{3.3}$$

and a specified initial and terminal condition

$$E_1 = E_0, \quad E_N \geq E_{final}. \tag{3.4}$$

The generator and battery are subject to power limits. In particular, the generator output satisfies

$$0 \leq P_t^g \leq P_{\max}^g, \tag{3.5}$$

while the battery charging and discharging power are bounded by

$$0 \leq P_t^c \leq P_{\max}^b, \quad 0 \leq P_t^d \leq P_{\max}^b. \tag{3.6}$$

The objective of the energy scheduling problem is to minimise the total operational cost over the scheduling horizon. This includes the cost of electricity purchased from the grid, generator fuel cost, and battery-related costs. Additional terms may be introduced in later sections to capture switching behaviour or discrete operational decisions.

The key parameters used in the system model are summarised in Table 3.1. These parameters are shared across all optimisation formulations to ensure a fair comparison.

Table 3.1: Model parameters and definitions

Symbol	**Unit**	**Description**
T	–	Set of time periods
Δt	h	Time step duration
D_t	kW	Electrical demand at time t
π_t	£/kWh	Electricity price at time t
$P_{\max}^{grid}$	kW	Maximum grid import capacity
$P_{\max}^g$	kW	Maximum generator output power
$P_{\min}^g$	kW	Minimum generator output when ON
c_{fuel}	£/kWh	Generator fuel cost
c_{on}	£/h	Generator fixed operating cost
c_{start}	£	Generator start-up cost
$P_{\max}^b$	kW	Maximum battery charge/discharge power
$P_{\min}^b$	kW	Minimum battery operating power (MIP)
η_c	–	Battery charging efficiency
η_d	–	Battery discharging efficiency
E_t	kWh	State of charge at time t
$E_{\min}$	kWh	Minimum battery energy level
$E_{\max}$	kWh	Maximum battery energy level
E_0	kWh	Initial state of charge
E_{final}	kWh	Required final state
c_c	£/kWh	Battery charging cost coefficient
c_d	£/kWh	Battery discharging cost coefficient
λ	–	Smoothing penalty weight
γ_{on}	–	Regularisation weight (generator usage)
β_{on}	–	Regularisation weight (switching)
γ_{Pb}	–	Regularisation weight (battery activity)

3.2 Model A: Convex Formulation

The convex formulation serves as a baseline model for microgrid energy scheduling, in which all decision variables are continuous. This formulation captures the fundamental energy flows and system constraints while maintaining computational tractability and global optimality.

The decision variables include grid import P_t^{grid}, generator output P_t^g, battery charging and discharging power P_t^c and P_t^d, and the battery state of charge E_t. All variables are continuous, and no binary or logical constraints are introduced.

The objective is to minimise the total operational cost over the scheduling horizon:

$$\min \sum_{t \in T} \left(\pi_t P_t^{grid} + c_{fuel} P_t^g + c_c P_t^c + c_d P_t^d \right) \Delta t + \lambda \sum_{t \in T,\, t \geq 2} (P_t^g - P_{t-1}^g)^2. \tag{3.7}$$

where the first term represents the energy cost and the second term is an optional quadratic smoothing penalty on generator output.

The convex formulation is subject to the shared system constraints introduced in Section 3.1, including power balance, battery dynamics, generator limits and battery power limits. Therefore, the model can be written compactly as minimising the objective above subject to (3.1)–(3.6).

When $\lambda = 0$, the model is a linear programme (LP). When $\lambda > 0$, it becomes a convex quadratic programme (QP). In both cases, the problem can be solved efficiently using standard solvers.

However, because all variables are continuous, this model cannot explicitly represent logical operational decisions such as generator on/off status, start-up events or mutually exclusive battery modes. These limitations motivate the regularised and mixed-integer formulations introduced in the following sections.

3.3 Model B: Regularised Convex

To bridge the gap between purely continuous optimisation and discrete decision-making, a regularised convex formulation is introduced. This approach augments the baseline convex model with additional penalty terms designed to promote structured behaviour, such as reduced generator usage and limited switching, without introducing binary variables.

A continuous auxiliary variable $s_t \in [0, 1]$ is introduced to represent a relaxed generator commitment level. The generator output is linked to this variable through

$$P_{\min}^g s_t \leq P_t^g \leq P_{\max}^g s_t, \quad \forall t \in T, \tag{3.8}$$

which enforces that generator output is zero when $s_t = 0$, and bounded within its operating range when $s_t > 0$.

To approximate discrete on/off behaviour, an ℓ_1-type penalty is applied to the relaxed commitment variable:

$$\gamma_{on} \sum_{t \in T} s_t, \tag{3.9}$$

which promotes sparsity in s_t and therefore encourages the generator to remain inactive for extended periods.

In addition, switching behaviour is approximated using a total variation (TV) penalty:

$$\beta_{on} \sum_{t \in T} |s_t - s_{t-1}|, \tag{3.10}$$

which discourages frequent changes in the generator state and promotes piecewise constant operation.

An optional ℓ_1 penalty is also introduced on the battery net power to reduce small oscillatory charge /discharge actions:

$$\gamma_{Pb} \sum_{t \in T} |P_t^d - P_t^c|. \tag{3.11}$$

The resulting optimisation problem can be written as

$$\begin{aligned} \min \quad & \sum_{t \in T} \left(\pi_t P_t^{grid} + c_{fuel} P_t^g + c_c P_t^c + c_d P_t^d \right) \Delta t \\ & + \gamma_{on} \sum_{t \in T} s_t + \beta_{on} \sum_{t \in T} |s_t - s_{t-1}| + \gamma_{Pb} \sum_{t \in T} |P_t^d - P_t^c| \\ & + \lambda \sum_{t \geq 2} (P_t^g - P_{t-1}^g)^2, \end{aligned} \tag{3.12}$$

subject to the shared constraints in Section 3.1, except that the continuous generator limit in (3.5) is replaced by the relaxed generator coupling constraint in (3.8).

This formulation remains convex, as all additional penalty terms are convex functions. As a result, the problem can still be solved efficiently while exhibiting behaviour that resembles discrete on/off scheduling.

However, it is important to note that this approach provides only a soft approximation of discrete decisions. While the regularisation terms can encourage sparsity and reduce switching, they do not enforce strict logical constraints such as minimum up/down times, exact start-up events, or hard operating thresholds. These limitations are addressed by the mixed-integer formulation in the next section.

3.4 Model C: Mixed-Integer

To explicitly represent discrete operational decisions, a mixed-integer formulation is introduced. In contrast to the previous models, binary variables are used to encode logical constraints such as generator commitment, start-up events, and battery operating modes.

A binary variable $z_t \in \{0, 1\}$ is introduced to represent the on/off status of the generator. The generator output is linked to this commitment variable through

$$P^g_{\min} z_t \leq P^g_t \leq P^g_{\max} z_t, \quad \forall t \in T, \tag{3.13}$$

which enforces that the generator is either completely off ($P^g_t = 0$ when $z_t = 0$) or operates within a feasible range when switched on.

To capture start-up events, an additional binary variable $y_t \in \{0, 1\}$ is introduced, defined by

$$y_t \geq z_t - z_{t-1}, \quad \forall t \in T, \tag{3.14}$$

with an initial condition relative to a given z_{init}. This ensures that $y_t = 1$ only when the generator transitions from an off state to an on state.

The battery operation is also modelled using binary variables to enforce mutually exclusive charging and discharging modes. Let $u^c_t, u^d_t \in \{0, 1\}$ denote the charging and discharging states, respectively. The following constraints apply:

$$u^c_t + u^d_t \leq 1, \quad \forall t \in T, \tag{3.15}$$

$$P^b_{\min} u^c_t \leq P^c_t \leq P^b_{\max} u^c_t, \quad P^b_{\min} u^d_t \leq P^d_t \leq P^b_{\max} u^d_t. \tag{3.16}$$

These constraints introduce a deadband, ensuring that the battery is either inactive or operates above a minimum power level when active.

The objective function extends the convex formulation by including fixed operating and start-up costs:

$$\begin{aligned} \min \quad & \sum_{t \in T} \left(\pi_t P^{grid}_t + c_{fuel} P^g_t + c_c P^c_t + c_d P^d_t \right) \Delta t \\ & + c_{on} \sum_{t \in T} z_t + c_{start} \sum_{t \in T} y_t + \lambda \sum_{t \geq 2} (P^g_t - P^g_{t-1})^2, \end{aligned} \tag{3.17}$$

This formulation results in a mixed-integer optimisation problem, which is generally non-convex and computationally more demanding than the convex and regularised models. However, it enables the explicit representation of logical and operational constraints, such as minimum generation levels, discrete start-up events, and mutually exclusive battery modes.

Therefore, the mixed-integer formulation serves as a reference model for capturing exact operational behaviour, against which the performance of the regularised convex approach can be evaluated.

3.5 Fair Comparison Protocol

To ensure a fair comparison, the three optimisation formulations are evaluated under the same experimental conditions. The system structure, parameter values, time horizon, time resolution and input datasets are fixed across all models. Therefore, any differences in the results are caused by the modelling formulation rather than by changes in data or system assumptions.

All models are implemented in AMPL and solved using the Gurobi optimisation solver on the same laptop/PC. The same solver settings are used where applicable, including the time limit and optimality tolerance. A standardised output format is adopted for all formulations, recording grid import, generator output, battery charge and discharge power, state of charge, generator state and switching indicators. This allows the MATLAB post-processing scripts to compute the same metrics and generate comparable figures for each model.

The formulations are also aligned in terms of operational purpose. The mixed-integer model represents generator commitment and start-up decisions using explicit binary variables and associated costs, while the regularised convex model uses sparsity and switching penalties to encourage similar behaviour without binary variables. This ensures that the comparison focuses on the role of discrete decision modelling, rather than on differences in the experimental setup.

3.6 Performance Metrics

To compare the three optimisation formulations, the evaluation metrics are divided into economic, behavioural and computational metrics.

Economic performance is measured using two quantities. The first is the raw operating cost, defined as

$$C_{\mathrm{op}} = \sum_{t \in T} \left(\pi_t P_t^{grid} + c_{fuel} P_t^{g} + c_c P_t^{c} + c_d P_t^{d} \right) \Delta t. \tag{3.18}$$

This cost excludes regularisation penalties, start-up costs and binary commitment penalties, and is therefore used as the common economic comparison across all models.

The second quantity is the model objective value. This includes the additional terms used by each formulation, such as regularisation penalties in the regularised convex model and start-up or on-cost terms in the mixed-integer model. It is used to explain why each model selects a particular dispatch pattern, but it is not used as the only measure of economic fairness.

Operational behaviour is evaluated using generator usage and switching activity. For the mixed-integer model, generator on-time and start-up events are measured by $\sum_t z_t$ and $\sum_t y_t$, respectively. For the regularised convex model, the corresponding surrogate measures are $\sum_t s_t$ and $\sum_t |s_t - s_{t-1}|$. Battery behaviour is assessed using the net battery power $P_t^b = P_t^d - P_t^c$ and its total variation.

Computational performance is assessed using solver runtime and, for the mixed-integer model, branch-and-bound information such as optimality gap and node count where available. These metrics allow the models to be compared in terms of cost, operational behaviour and computational complexity.

4 Results: Synthetic Benchmark

4.1 Data Description

A synthetic benchmark dataset is generated to compare the convex, regularised convex, and mixed-integer optimisation models under controlled conditions. The scheduling horizon is 168 hours with an hourly time resolution.Both demand and electricity price are synthetically generated time-varying profiles. The load follows a daily pattern with lower night-time demand and morning/evening peaks, while the price follows a time-of-use structure with higher evening prices. Since these inputs vary with time, their full profiles are shown in Appendix B.3 rather than listed as fixed parameters in Table 4.1.The key fixed system parameters used in the benchmark are summarised in Table 4.1.

Table 4.1: System parameters used in the synthetic benchmark

Parameter	Description	Value
N	Scheduling horizon	168 h
Δt	Time resolution	1 h
$P_{\text{grid}}^{\max}$	Grid import limit	12 kW
$P_g^{\max}$	Generator maximum output	5 kW
$P_g^{\min}$	Generator minimum output	0.05 kW
c_{fuel}	Generator fuel cost	£0.10 /kWh
$P_b^{\max}$	Battery power limit	5 kW
$E_{\min}$	Minimum battery energy	2 kWh
$E_{\max}$	Maximum battery energy	10 kWh
E_0	Initial state of charge	5 kWh
E_{final}	Final SoC requirement	5 kWh
η_c	Charging efficiency	0.95
η_d	Discharging efficiency	0.95
c_{on}	Generator on cost	£0.02/h
c_{start}	Generator start cost	£0.02
γ_{on}	Regularisation weight (on-time)	0.02
β_{on}	Regularisation weight (switching)	0.02
γ_{P_b}	Battery power penalty	0.005

4.2 Dispatch Comparison

The generator dispatch profiles produced by the three optimisation formulations are shown in Fig. 4.1. The generator is mainly used during periods where local generation becomes economically attractive compared with grid import. These periods correspond to the higher-price intervals in the synthetic electricity price profile shown in Appendix B.3.

The convex formulation produces several short and intermittent generator outputs because it has no explicit commitment variable or start-up cost. As a result, the generator can be used for small adjustments whenever this slightly reduces the energy cost. In contrast, the mixed-integer formulation activates the generator only during a few distinct periods, because the binary commitment constraint and start-up cost make short or unnecessary operation less attractive.

The regularised convex model gives an intermediate result. The sparsity and switching penalties reduce unnecessary generator use and discourage frequent changes, so its dispatch pattern becomes closer to the mixed-integer solution while still remaining continuous. Overall, Fig. 4.1 shows that the three models respond to the same load and price signals, but differ in how strongly they penalise small or frequent generator operation.

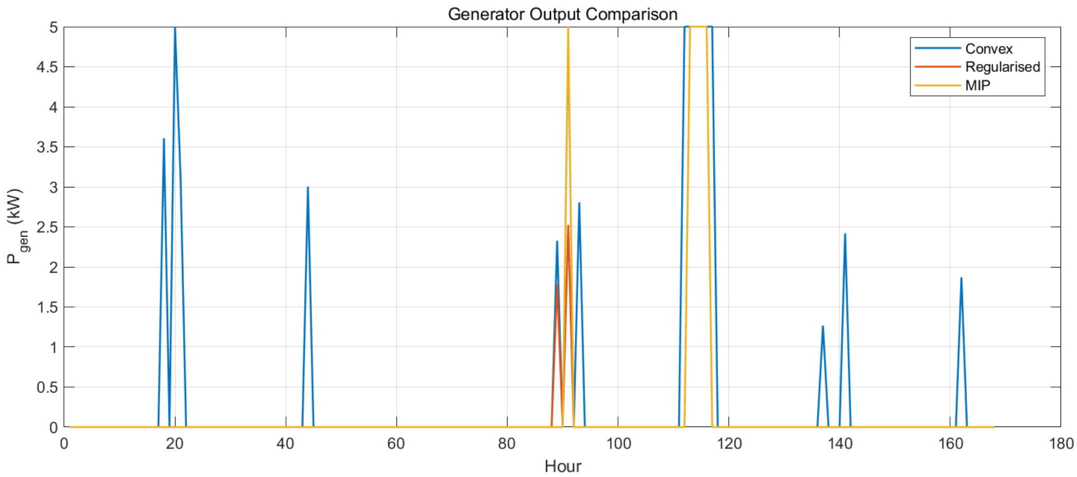

Figure 4.1: Generator output comparison between the convex, regularised convex, and mixed-integer models.

The grid import behaviour under the three optimisation models is shown in Fig. 4.2. This figure is more variable than the generator dispatch because grid import acts as the balancing source in the system: it supplies the remaining demand after generator output and battery charging or discharging have been considered.

The main feature to observe is the inverse relationship between grid import and local supply. When the generator or battery discharge contributes more power, the required grid import decreases. Conversely, when the generator is off and the battery is charging or unable to discharge, grid import increases to satisfy the load. Therefore, the sharp variations in Fig. 4.2 mainly reflect the combined effects of the time-varying load, electricity price, and battery operation.

Compared with the convex formulation, the mixed-integer and regularised models show more structured grid-import behaviour because generator use is penalised through commitment, start-up, sparsity or switching terms. Overall, Fig. 4.2 should be interpreted together with the generator and battery profiles, since grid import represents the residual power required from the main grid.

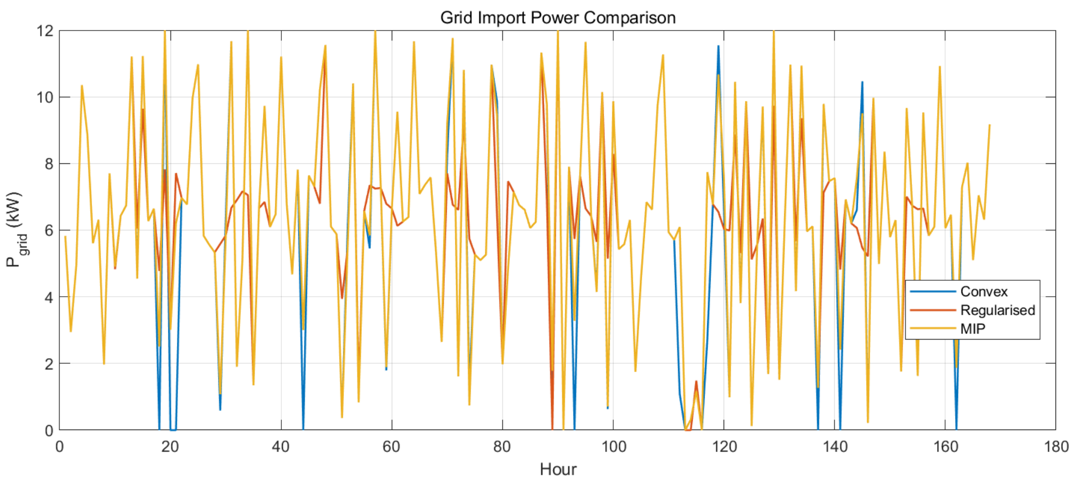

Figure 4.2: Grid import power comparison under the synthetic benchmark scenario.

Figure 4.3 compares the battery charging and discharging behaviour across the three models. Positive values correspond to battery discharge, while negative values indicate charging. The battery is actively used in all three models to shift energy across time in response to electricity price variations. The overall battery behaviour is broadly similar across the three formulations, indicating that battery operation is primarily driven by price signals rather than generator commitment decisions.

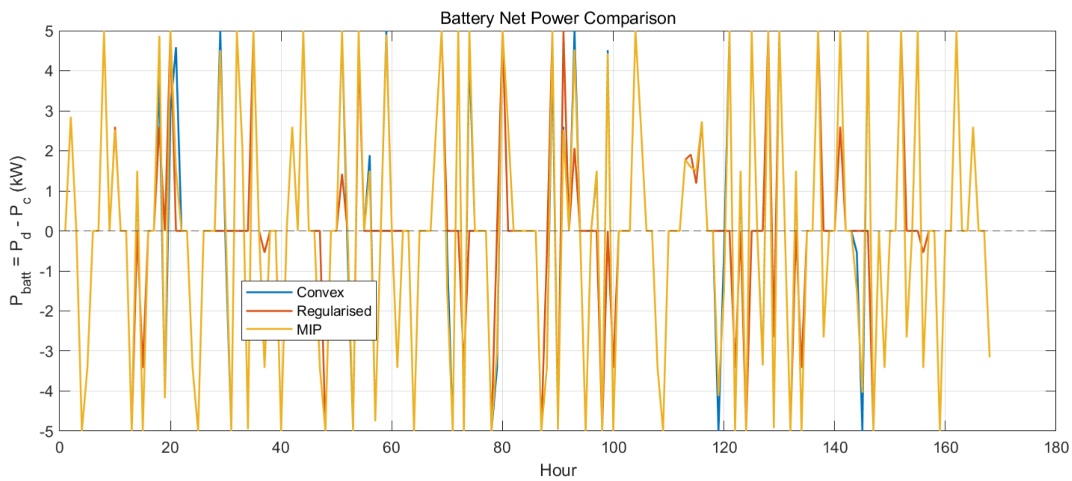

Figure 4.3: Battery net power comparison for the three optimisation formulations.

4.3 Battery SoC Comparison

Figure 4.3 illustrates the battery net power profile across the scheduling horizon. Positive values indicate discharge and negative values indicate charging. The battery is actively used in all three models to shift energy over time, with broadly similar behaviour observed across the convex, regularised convex, and mixed-integer formulations.

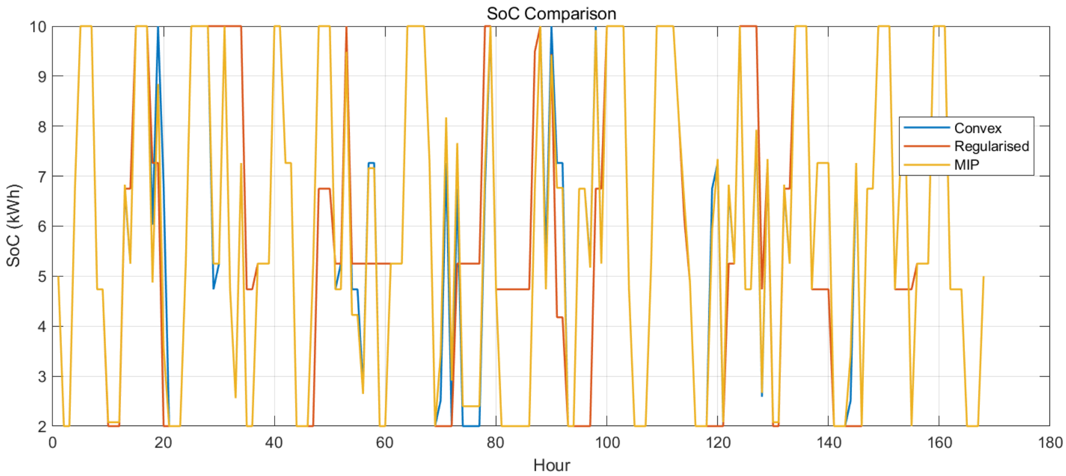

Figure 4.4: Battery state of charge comparison.

4.4 Cost Comparison

Table 4.2 summarises the raw operating cost obtained from the three optimisation formulations and the grid-only baseline. The raw operating cost includes only the common economic terms: grid energy cost, generator fuel cost, and battery charging/discharging costs. It excludes formulation-specific penalties such as regularisation terms, generator on-costs and start-up costs, so that the models can be compared on the same economic basis.

All optimisation models achieve lower raw operating costs than the grid-only baseline, showing that coordinated generator and battery scheduling can reduce the cost of supplying the load. The convex formulation gives the lowest raw operating cost because it has the fewest operational restrictions. It can use small and intermittent generator outputs whenever they slightly reduce the energy cost, so this result should be interpreted as an optimistic lower-cost benchmark rather than the most realistic operating strategy.

The mixed-integer model gives a slightly higher raw operating cost because it includes binary commitment and start-up logic. These constraints prevent the generator from being used freely at arbitrarily small levels and therefore capture more realistic operational behaviour. The regularised convex model gives the highest raw operating cost among the optimisation models because its sparsity and switching penalties discourage unnecessary generator use and frequent changes. Therefore, the higher costs of the regularised and mixed-integer models reflect the economic cost of representing more structured and operationally realistic behaviour.

Table 4.2: Total operating cost comparison for the synthetic benchmark

Model	Total Cost (£)
Grid-only baseline	87.95
Convex optimisation	83.76
Regularised convex optimisation	85.49
Mixed-integer optimisation	84.03

4.5 Discussion

The results presented in the previous sections show that the total operating costs obtained from the three optimisation formulations are relatively close. This outcome can be explained by the overall dispatch patterns observed in the system. As shown in the generator dispatch profiles, the generator is activated only during a few periods of the scheduling horizon. Consequently, the influence of generator commitment decisions on the total cost remains limited.

In addition, the battery net power profiles indicate that the battery is actively used in all formulations to shift energy across time in response to electricity price variations. Since the battery behaviour is broadly similar across the convex, regularised convex, and mixed-integer models, the resulting system operation remains comparable.

Therefore, although the mathematical formulations differ, the overall dispatch patterns remain similar, which leads to only small differences in total operating cost across the three optimisation approaches.

5 Results: Parameter Sweep

5.1 Penalty Parameter Setup

The regularised convex formulation introduces penalty terms to encourage sparse generator usage and reduce excessive switching. Two penalty parameters are considered: an on-time penalty and a switching penalty.

The generator on-time penalty γ_{on} penalises the relaxed commitment variable s_t,

$$\gamma_{\text{on}} \sum_{t \in T} s_t, \tag{5.1}$$

which increases the cost of keeping the generator active and therefore promotes sparse generator utilisation.

A switching penalty β_{on} is also introduced,

$$\beta_{\text{on}} \sum_{t \in T} |s_t - s_{t-1}|, \tag{5.2}$$

which discourages frequent changes in generator operation and approximates the effect of start-up costs used in mixed-integer formulations.

To analyse the sensitivity of the model to these penalties, logarithmic parameter sweeps were performed. The base values of the penalties were scaled using

$$k, r \in \{0.001,\ 0.003,\ 0.01,\ 0.03,\ 0.1,\ 0.3,\ 1,\ 3,\ 10,\ 100,\ 1000\}. \tag{5.3}$$

The effective parameters were defined as

$$\gamma_{\text{on}}^{(k)} = k\,\gamma_{\text{on}}^{\text{base}}, \qquad \beta_{\text{on}}^{(r)} = r\,\beta_{\text{on}}^{\text{base}}. \tag{5.4}$$

For each parameter value, the optimisation model was solved using the same 168-hour benchmark dataset. The resulting solutions were analysed in terms of generator utilisation, switching behaviour, and operating cost, as discussed in the following section.

5.2 Trade-off Analysis

To evaluate the behaviour of the regularised convex formulation, parameter sweeps were performed for the generator usage penalty γ and the switching penalty β. Each parameter was varied across several orders of magnitude, and the resulting solutions were analysed using metrics such as total cost, generator utilisation and switching behaviour.

Figure 5.1 compares total generator energy and total cost for different values of γ. The results show that increasing γ does not lead to a strictly monotonic increase in cost. Intermediate values of γ reduce generator usage while producing a higher objective value, whereas very large γ almost eliminates generator operation and gives a relatively low cost. This indicates that γ changes the dispatch structure rather than simply increasing cost.

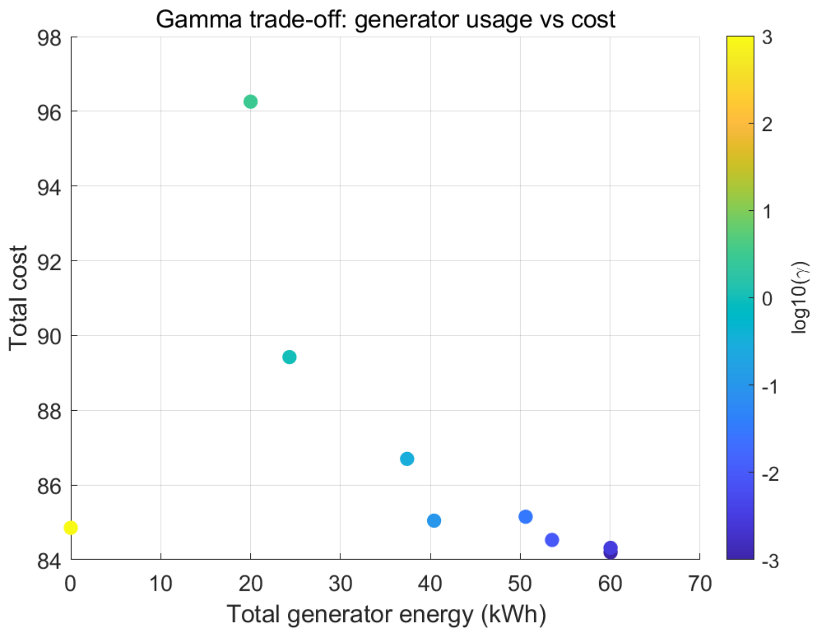

Figure 5.1: Trade-off between generator usage and total cost for different values of γ.

Figure 5.2 compares the regularised solutions with the MIP reference using the L1 distance of generator output. The results show no simple monotonic relationship between similarity to MIP and total cost. Although some values of γ make the regularised dispatch closer to the MIP solution, they also lead to a higher objective value. This suggests that γ can influence the generator pattern, but it cannot reliably reproduce the MIP solution by itself.

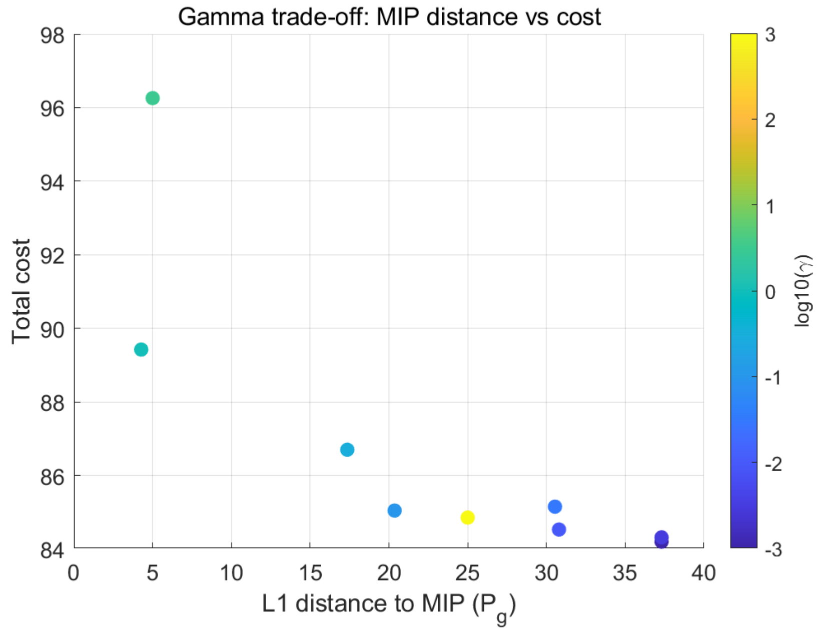

Figure 5.2: Relationship between distance to the mixed-integer solution and total cost for different values of γ.

Figure 5.3 illustrates the effect of the switching penalty β. Larger values of β generally reduce the switching metric $\sum|\Delta s_g|$, showing that the penalty encourages smoother generator operation. However, the total cost is non-monotonic, with some intermediate β values leading to higher objective values. Therefore, β is effective in reducing switching, but it does not guarantee a lower cost.

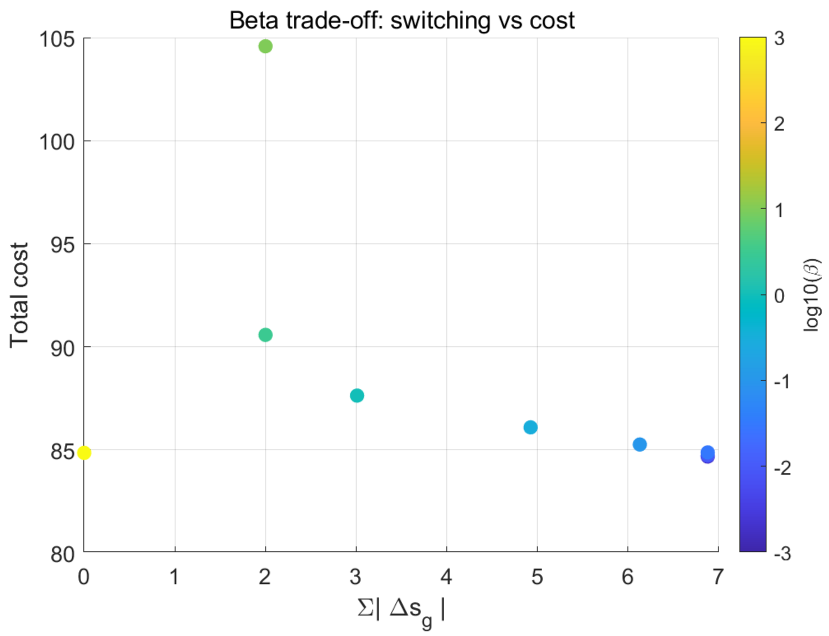

Figure 5.3: Trade-off between switching activity and total cost for different values of β.

Figure **??** shows that the relationship between β and similarity to the MIP solution is non-monotonic. Moderate values of β can reduce the L1 distance to the MIP generator profile, but they may also increase the objective value. Very large β over-smooths the generator schedule, leading to a larger distance from the MIP solution. Therefore, β helps control switching, but it does not guarantee a closer match to the MIP dispatch.

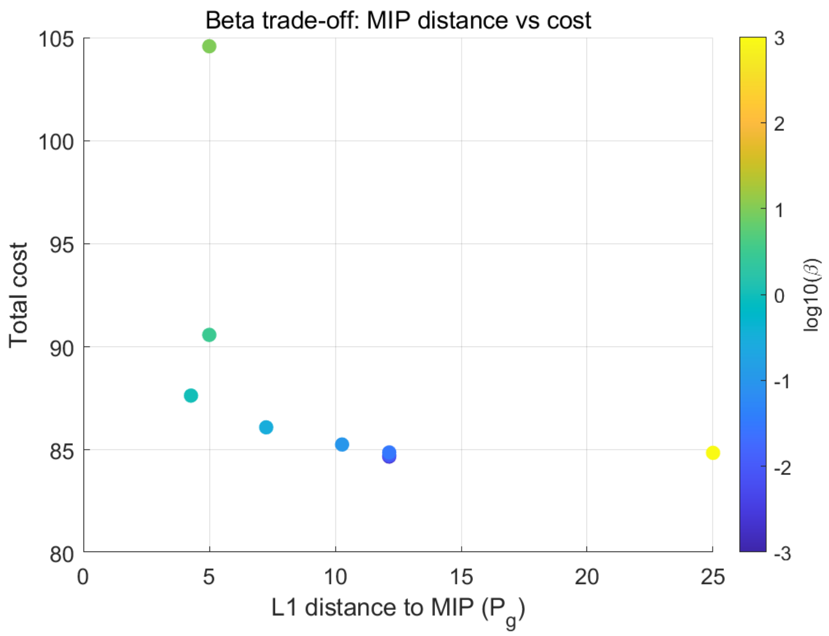

Figure 5.4: Relationship between distance to the mixed-integer solution and total cost for different values of β.

5.3 When Convex Matches MIP

The parameter sweep results indicate that the regularised convex model can approximate some mixed-integer behaviours, but only within a suitable range of penalty parameters. The effects of both γ and β are non-monotonic. Increasing γ can reduce generator usage, while increasing β can reduce switching activity. However, neither parameter guarantees a closer match to the MIP solution.

For some intermediate penalty values, the regularised model produces generator schedules that are closer to the MIP reference, although this may come with a higher objective value. In contrast, very large penalties can over-penalise generator use or over-smooth the schedule, leading to dispatch patterns that differ from the MIP solution.

Therefore, the regularised convex formulation can reproduce MIP-like behaviour in a soft sense, especially sparse generator operation and reduced switching. However, it cannot enforce exact logical constraints such as binary commitment, start-up events, or minimum operating thresholds. Binary variables remain necessary when these operational decisions must be represented exactly.

6 Results: UK National Grid Data

6.1 Data Source

To evaluate the optimisation models under realistic operating conditions, UK wholesale electricity price data were obtained from the Elexon Balancing Mechanism Reporting Service (BMRS) market index price database[1].

The selected dataset covers one week, from 15 February 2026 to 21 February 2026, giving a 168-hour scheduling horizon. The raw price data were processed and converted to an hourly time series to match the optimisation model resolution. The resulting UK electricity price profile is shown in Appendix B.3.

The same demand profile used in the synthetic benchmark is retained in this experiment. This allows the effect of realistic electricity price variations to be isolated while keeping the demand conditions consistent across the synthetic and UK-data case studies.

6.2 Model Performance on Real Data

The three optimisation models were evaluated using the UK electricity price dataset described in Section 6.1. The resulting operational behaviour of the microgrid is analysed by comparing generator output, grid import, and battery operation across the convex, regularised convex, and mixed-integer formulations.

Figure 6.1 shows the generator output obtained from the three models. The mixed-integer model activates the generator only during a limited number of hours when electricity prices are relatively high, producing a clear on/off operational pattern. The regularised convex formulation produces a very similar behaviour, where the relaxed commitment variable s_g closely approximates the binary generator state. In contrast, the baseline convex model occasionally produces small generator outputs due to the absence of explicit commitment constraints.

The corresponding generator commitment behaviour is illustrated in Figure 6.2. The relaxed commitment variable produced by the regularised convex model closely follows the binary commitment decision of the mixed-integer formulation, indicating that the regularisation terms successfully approximate discrete operational decisions.

Figure 6.3 compares the grid import power for the three models. The overall trends are similar because

[1] `https://bmrs.elexon.co.uk/market-index-prices`

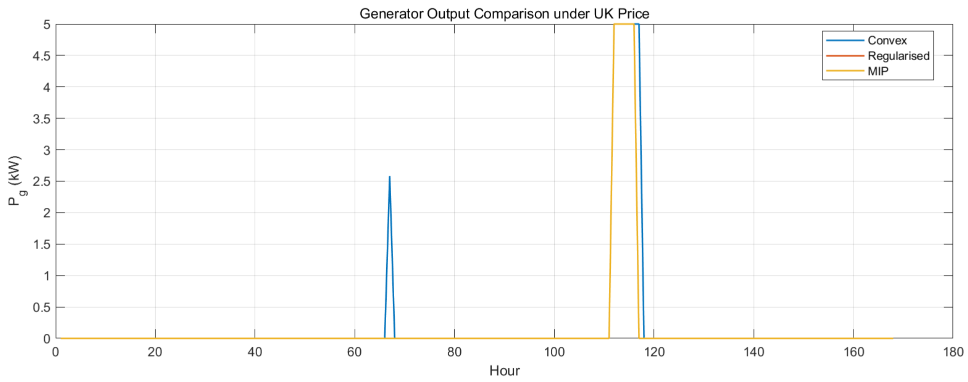

Figure 6.1: Generator output comparison under UK electricity prices.

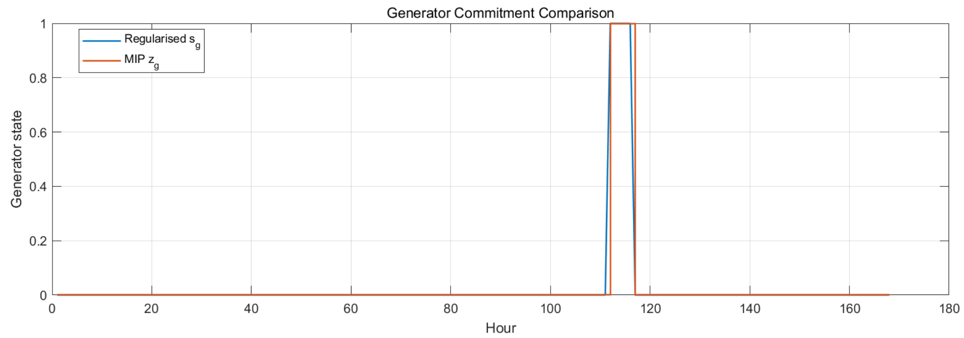

Figure 6.2: Generator commitment comparison between regularised and MIP models.

grid import balances the remaining demand after generator and battery contributions. However, the convex model occasionally exhibits sharper variations due to its fully continuous dispatch decisions, while the mixed-integer and regularised formulations produce slightly more structured behaviour.

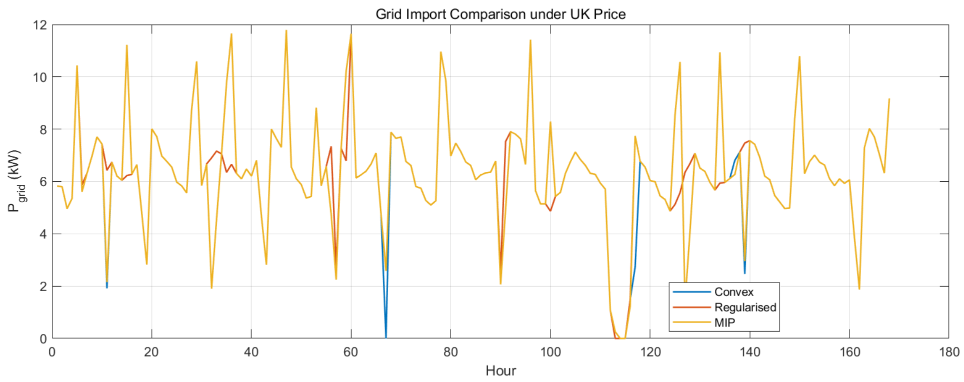

Figure 6.3: Grid import comparison under UK electricity prices.

Battery operation is shown in Figure 6.4, while the corresponding state-of-charge trajectories are presented in Figure 6.5. All models demonstrate price-driven battery operation, where the battery charges during lower-price periods and discharges during higher-price intervals. The mixed-integer model produces clearer charging and discharging segments due to its explicit mode constraints. The regularised convex model closely reproduces this behaviour, whereas the convex baseline occasionally exhibits small intermediate power levels.

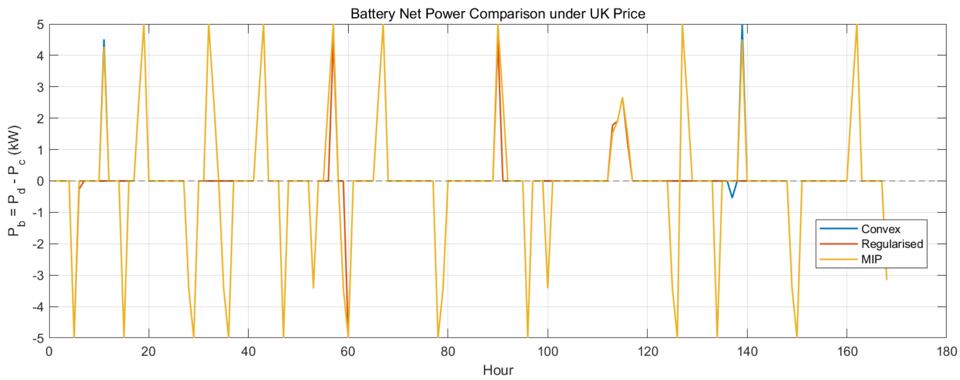

Figure 6.4: Battery net power comparison under UK electricity prices.

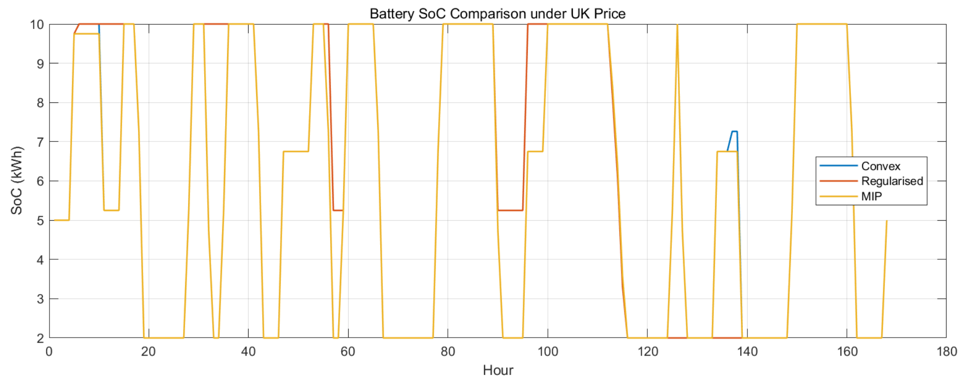

Figure 6.5: Battery state-of-charge comparison under UK electricity prices.

Overall, the results obtained with real electricity price data are consistent with the findings from the synthetic benchmark experiments. The mixed-integer model provides an exact representation of discrete operational logic, while the regularised convex formulation is capable of reproducing similar operational patterns through sparsity and switching penalties. The convex baseline remains computationally efficient but cannot enforce strict commitment behaviour.

6.3 Discussion

The UK-data results are consistent with the synthetic benchmark. The mixed-integer model gives the clearest representation of discrete operating logic through explicit generator commitment and start-up decisions. The regularised convex model approximates this behaviour using sparsity and switching penalties, while the baseline convex model remains simpler but less structured.

Overall, the UK-data case confirms the main trade-off: mixed-integer optimisation provides higher modelling fidelity, whereas regularised convex optimisation offers a cheaper continuous approximation that can reproduce many key operational features without binary variables.

7 Conclusions and Future Work

7.1 Summary of Findings

This dissertation compared three optimisation formulations for microgrid energy scheduling with battery storage: a convex baseline model, a regularised convex model, and a mixed-integer model. A unified comparison framework was adopted so that all models were evaluated under the same system parameters, time horizon, solver settings, and input data.

Table 7.1 summarises the total cost results for the two main case studies.

Table 7.1: Comparison of total cost across optimisation formulations

Formulation	Experiment 1/Synthetic Baseline	Experiment 3/Nation Grid Data
Convex	83.76	82.99
Regularised convex	85.50	83.79
Mixed-integer	84.03	83.11
Maximum benchmark	87.95	85.33

The results show that the total cost differences between the three formulations were small in both case studies. In Experiment 1, the spread between the lowest and highest cost was less than 2 units, while in Experiment 3 it was below 1 unit. This indicates that, under the selected data and parameter settings, all three models achieved broadly similar economic performance.

One possible explanation is that the same demand, price signals, and physical limits were imposed across all formulations, leading to similar dispatch decisions overall. Although the mixed-integer and regularised convex models imposed stronger structure on generator commitment and battery behaviour, these differences mainly influenced the operating pattern rather than the final total cost.

The convex model remained the simplest formulation and achieved the lowest cost in both experiments, but it could not explicitly represent discrete operational logic. The regularised convex model reproduced some important mixed-integer features while preserving a continuous formulation, whereas the mixed-integer model provided the clearest representation of generator commitment and battery operating logic at the cost of higher modelling complexity.

If larger cost differences are to be observed more clearly, future experiments could vary parameters that increase the importance of commitment decisions, such as generator start-up cost, minimum generator output, battery degradation penalty, or electricity price volatility.

Overall, the study shows that regularised convex optimisation can provide a useful approximation to mixed-integer behaviour when cost differences are modest, whereas mixed-integer optimisation remains necessary when exact discrete operating decisions must be enforced.

7.2 When Binary Variables Are Necessary

The experiments show that binary variables are not always required to achieve similar total-cost performance. In both case studies, the cost differences between the three formulations were small, suggesting that continuous or regularised models can already provide reasonable scheduling behaviour in some cases.

However, binary variables become necessary when hard operational logic must be represented exactly, including generator commitment, start-up events, minimum generation thresholds, and mutually exclusive battery charging and discharging modes. While the regularised convex model could reproduce some mixed-integer features, such as reduced switching and sparser generator use, it could not guarantee exact discrete decisions. Therefore, mixed-integer optimisation remains necessary when accurate representation of operational logic is more important than the modest cost differences observed in this study.

7.3 Limitations

This study has several limitations. The microgrid model is intentionally simplified, including only a generator, battery, grid connection, and fixed demand profile. The formulation is deterministic and does not consider renewable uncertainty, forecast errors, or unexpected disturbances. Network constraints, such as power-flow effects and transmission limits, are also omitted. In addition, the real-data validation is limited to one 168-hour UK price dataset. These simplifications are useful for isolating modelling differences, but they limit the generality of the conclusions.

7.4 Future Work

Future work could include renewable generation, uncertainty, and larger networked microgrids, where the differences between convex and mixed-integer formulations may become more pronounced. More detailed battery degradation models, additional generator constraints, and longer-horizon or real-time control could also be considered.

8 Economic, Legal, Social, Ethical and Environmental Context

8.1 Economic Considerations

This project is closely related to the economic operation of microgrids, as the optimisation framework aims to reduce operating cost through coordinated scheduling of grid import, generator output, and battery usage. The comparison is also economically relevant because it highlights the trade-off between solution quality and computational effort: convex models are cheaper to solve, while mixed-integer models can represent operational logic more accurately at higher computational cost.

8.2 Legal and Regulatory Context

Although this dissertation focuses on methodological comparison, practical microgrid scheduling must comply with grid connection rules, electricity market arrangements, and equipment operating limits. In this study, the simplified assumption of grid import only avoids additional regulatory issues associated with electricity export and market participation, but these would become important in real deployment.

8.3 Environmental Impact

The project has environmental relevance because improved scheduling can reduce unnecessary generator use and support more efficient energy management. However, cost minimisation does not always imply lower emissions, since a generator may still be used when it is economically favourable. Future work could therefore include emission-related objectives or constraints.

8.4 Ethical Considerations

From an ethical perspective, optimisation results should be applied with caution and should support, rather than replace, engineering judgement. Simplified models may omit practical constraints, so transparency and interpretability are important when using optimisation tools in energy system decision-making.

Reference

[1] D. E. Olivares et al., "Trends in microgrid control," *IEEE Transactions on Smart Grid*, vol. 5, no. 4, pp. 1905–1919, 2014.

[2] E. Mengelkamp, J. Gärttner, K. Rock, S. Kessler, L. Orsini, and C. Weinhardt, "Designing microgrid energy markets: A case study," *Applied Energy*, vol. 210, pp. 870–880, 2018.

[3] B. Zakeri and S. Syri, "Electrical energy storage systems: A comparative life cycle cost analysis," *Renewable and Sustainable Energy Reviews*, vol. 42, pp. 569–596, 2015.

[4] H. Chen, T. N. Cong, W. Yang, C. Tan, Y. Li, and Y. Ding, "A review of battery energy storage systems for grid applications," *Renewable and Sustainable Energy Reviews*, vol. 120, p. 109 670, 2020.

[5] Y. Zhang, N. Gatsis, and G. B. Giannakis, "Review on optimal energy management of microgrids," *IEEE Transactions on Smart Grid*, vol. 9, no. 5, pp. 1–13, 2018.

[6] J. M. Morales, A. J. Conejo, H. Madsen, P. Pinson, and M. Zugno, *Integrating Renewables in Electricity Markets*. Springer, 2017.

[7] N. Li, L. Chen, and S. H. Low, "Convex relaxation of optimal power flow and its applications," *Proceedings of the IEEE*, vol. 107, no. 11, pp. 1–20, 2019.

[8] Y. Xu and W. Zhang, "Data-driven approaches for power system optimization," *IEEE Transactions on Power Systems*, vol. 35, no. 1, pp. 1–10, 2020.

[9] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.

[10] X. Liu, T. Zhao, H. Deng, P. Wang, J. Liu, and F. Blaabjerg, "Microgrid energy management with energy storage systems: A review," *CSEE Journal of Power and Energy Systems*, vol. 9, no. 2, pp. 483–504, 2023. DOI: 10.17775/CSEEJPES.2022.04290.

[11] J. Vaičys et al., "A case study of optimising energy storage dispatch: Convex optimisation approach with degradation considerations," *HardwareX*, vol. 18, e00531, 2024. DOI: 10.1016/j.ohx.2024.e00531.

[12] G. S. Thirunavukkarasu, M. Seyedmahmoudian, E. Jamei, B. Horan, S. Mekhilef, and A. Stojcevski, "Role of optimization techniques in microgrid energy management systems—a review," *Energy Strategy Reviews*, vol. 43, p. 100 899, 2022. DOI: 10.1016/j.esr.2022.100899.

[13] S. E. Eyimaya and N. Altin, "Review of energy management systems in microgrids," *Applied Sciences*, vol. 14, no. 3, p. 1249, 2024. DOI: 10.3390/app14031249.

[14] N. Q. Minh et al., “A mixed-integer linear programming model for microgrid energy management considering battery degradation cost,” *Sustainable Energy, Grids and Networks*, vol. 39, p. 101 434, 2024.

[15] G. Liu et al., “A mixed integer linear programming-based distributed energy scheduling method,” *IET Energy Systems Integration*, 2023. DOI: 10.1049/esi2.12103.

[16] L. A. Roald, D. Pozo, A. Papavasiliou, D. K. Molzahn, J. Kazempour, and A. J. Conejo, “Power systems optimization under uncertainty: A review of methods and applications,” *Electric Power Systems Research*, vol. 214, p. 108 725, 2023. DOI: 10.1016/j.epsr.2022.108725.

[17] A. R. Abbasi et al., “Recent developments of energy management strategies in microgrids: A review,” *Energy Conversion and Management*, vol. 287, p. 117 042, 2023.

[18] M. F. Zia, E. H. Elbouchikhi, M. Benbouzid, and J. M. Guerrero, “Microgrids energy management systems: A critical review on methods, solutions, and prospects,” *Applied Energy*, vol. 222, pp. 1033–1055, 2018. DOI: 10.1016/j.apenergy.2018.04.103.

[19] Y. Wang, N. Zhang, and C. Kang, “Energy management of microgrids using advanced optimization techniques: A review,” *Applied Energy*, vol. 335, p. 120 705, 2023. DOI: 10.1016/j.apenergy.2023.120705.

[20] D. Tenfen and E. C. Finardi, “A mixed integer linear programming model for the energy management problem of microgrids,” *Electric Power Systems Research*, vol. 122, pp. 19–28, 2015. DOI: 10.1016/j.epsr.2014.12.019.

[21] S. W. Park et al., “A comprehensive review of battery-based power service strategy and degradation modeling for electric power system planning and operation,” *Applied Energy*, vol. 367, p. 123 680, 2024.

[22] Y. Zhang, X. Li, and J. Wang, “Battery degradation modeling and optimization for microgrid energy management,” *Journal of Energy Storage*, vol. 51, p. 104 458, 2022. DOI: 10.1016/j.est.2022.104458.

[23] C. Álvarez-Arroyo et al., “Optimising microgrid energy management: Leveraging energy storage systems for enhanced stochastic scheduling under uncertainty,” *Renewable Energy*, vol. 236, p. 121 674, 2024.

[24] W. Dong et al., “Forecast-driven stochastic optimization scheduling of an energy management system for an isolated hydrogen production microgrid,” *Energy*, vol. 263, p. 125 784, 2023.

[25] D. Bertsimas, R. Cory-Wright, and J. Pauphilet, “A unified approach to mixed-integer optimization problems with logical constraints,” *SIAM Journal on Optimization*, vol. 31, no. 3, pp. 2340–2367, 2021. DOI: 10.1137/20M1346778.

[26] S. Fattahi, M. Ashraphijuo, J. Lavaei, and A. Atamtürk, “Conic relaxations of the unit commitment problem,” *Energy*, vol. 134, pp. 1079–1095, 2017. DOI: 10.1016/j.energy.2017.06.121.

[27] P. O. Kriett and M. Salani, "Optimal control of a residential microgrid," *Energy*, vol. 42, no. 1, pp. 321–330, 2012. DOI: 10.1016/j.energy.2012.03.049.

[28] T. Nguyen-Duc, L. Hoang-Tuan, H. Ta-Xuan, L. Do-Van, and H. Takano, "A mixed-integer programming approach for unit commitment in micro-grid with incentive-based demand response and battery energy storage system," *Energies*, vol. 15, no. 19, p. 7192, 2022. DOI: 10.3390/en15197192.

Appendix

A Model Formulation Details

This appendix provides the full mathematical details of the optimisation models used in Chapter 3. It supplements the main methodology chapter by collecting the modelling assumptions, notation, shared system constraints, and complete optimisation formulations in one place.

Figure 1 shows the grid-connected microgrid structure used in this dissertation. The system consists of a main grid, a dispatchable generator, a battery energy storage system (BESS), and a fixed load demand. The grid is assumed to be always available and electricity can be imported at time-varying prices. The generator provides local dispatchable supply, while the battery charges during low-price periods and discharges during high-price periods.

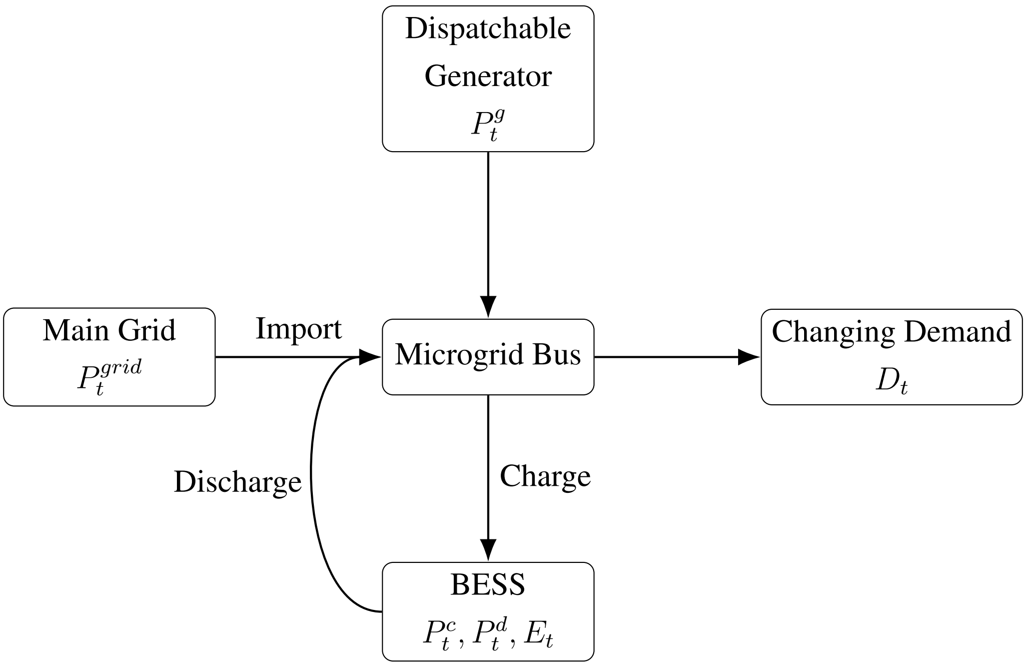

Figure 1: Schematic diagram of the grid-connected microgrid system.

A.1 System Assumptions

The microgrid is modelled as a single-bus system consisting of grid import, a dispatchable generator, a battery energy storage system, and a fixed electrical demand. The following assumptions are adopted throughout the dissertation:

a) The system is represented by a single electrical bus, so network losses and power-flow constraints are neglected.

b) The scheduling horizon is discretised into equal time intervals of length Δt.

c) The demand profile D_t is deterministic and known in advance over the optimisation horizon.

d) The main grid is always available.

e) The generator has a linear fuel cost and a bounded operating range.

f) Battery charging and discharging efficiencies are constant.

g) The battery state of charge follows a linear discrete-time energy balance equation.

h) The same physical system parameters are used across all formulations in order to ensure a fair comparison.

A.2 Notation and Parameters

Table 1 summarises the main symbols used in the model formulations.

Table 1: Model parameters and decision variables used in the optimisation formulations.

Symbol	**Unit**	**Description**
T	–	Set of time periods
Δt	h	Time-step duration
D_t	kW	Electrical demand at time t
π_t	£/kWh	Grid electricity price at time t
$P^{grid}_{\max}$	kW	Maximum grid import power
$P^{g}_{\max}$	kW	Maximum generator output power
$P^{g}_{\min}$	kW	Minimum generator output when committed
$P^{b}_{\max}$	kW	Maximum battery charging/discharging power
$P^{b}_{\min}$	kW	Minimum battery operating power in the mixed-integer model
η_c	–	Battery charging efficiency
η_d	–	Battery discharging efficiency
$E_{\min}$	kWh	Minimum state of charge
$E_{\max}$	kWh	Maximum state of charge
E_0	kWh	Initial battery state of charge
E_{final}	kWh	Required final state of charge
c_{fuel}	£/kWh	Generator fuel cost coefficient
c_c	£/kWh	Battery charging cost coefficient
c_d	£/kWh	Battery discharging cost coefficient
c_{on}	£/h	Generator fixed operating cost in the mixed-integer model
c_{start}	£	Generator start-up cost in the mixed-integer model
λ	–	Quadratic smoothing penalty weight
γ_{on}	–	Generator-usage regularisation weight
β_{on}	–	Switching regularisation weight
γ_{Pb}	–	Battery activity regularisation weight
P^{grid}_t	kW	Grid import power at time t
P^{g}_t	kW	Generator output power at time t
P^{c}_t	kW	Battery charging power at time t
P^{d}_t	kW	Battery discharging power at time t
E_t	kWh	Battery state of charge at time t
s_t	–	Relaxed generator commitment variable in the regularised model
z_t	–	Binary generator commitment variable in the mixed-integer model
y_t	–	Binary generator start-up variable in the mixed-integer model
u^c_t	–	Binary battery charging-mode variable
u^d_t	–	Binary battery discharging-mode variable

A.3 Shared System Constraints

The three formulations use the same physical model of the microgrid. The common constraints are listed below.

Power balance

For each time period, total supply must match demand and battery charging:

$$P_t^{grid} + P_t^g + P_t^d = D_t + P_t^c, \qquad \forall t \in T. \tag{1}$$

Grid import limit

The imported grid power is bounded by the grid connection capacity:

$$0 \leq P_t^{grid} \leq P_{\max}^{grid}, \qquad \forall t \in T. \tag{2}$$

Battery power limits

Battery charging and discharging powers are individually bounded:

$$0 \leq P_t^c \leq P_{\max}^b, \qquad \forall t \in T, \tag{3}$$

$$0 \leq P_t^d \leq P_{\max}^b, \qquad \forall t \in T. \tag{4}$$

Battery state-of-charge dynamics

The battery energy evolves according to the discrete-time balance equation:

$$E_1 = E_0 + \eta_c P_1^c \Delta t - \frac{1}{\eta_d} P_1^d \Delta t, \tag{5}$$

$$E_t = E_{t-1} + \eta_c P_t^c \Delta t - \frac{1}{\eta_d} P_t^d \Delta t, \qquad \forall t \in T,\ t \geq 2. \tag{6}$$

State-of-charge bounds

The battery state of charge must remain within its operating range:

$$E_{\min} \leq E_t \leq E_{\max}, \qquad \forall t \in T. \tag{7}$$

Terminal state-of-charge requirement

A minimum final energy level is imposed:

$$E_{|T|} \geq E_{\text{final}}. \tag{8}$$

A.4 Model A: Convex Formulation

The baseline convex model uses only continuous decision variables. Its objective is to minimise the total operating cost over the scheduling horizon:

$$\min \sum_{t \in T} \left(\pi_t P_t^{grid} + c_{\text{fuel}} P_t^g + c_c P_t^c + c_d P_t^d \right) \Delta t + \lambda \sum_{t \in T,\, t \geq 2} \left(P_t^g - P_{t-1}^g \right)^2. \tag{9}$$

The generator output is bounded by

$$0 \leq P_t^g \leq P_{\max}^g, \qquad \forall t \in T. \tag{10}$$

Together with (1)–(8), this gives a convex optimisation problem. When $\lambda = 0$, the formulation reduces to a linear programme; when $\lambda > 0$, it becomes a convex quadratic programme.

A.5 Model B: Regularised Convex Formulation

The regularised convex model introduces a continuous surrogate variable $s_t \in [0, 1]$ to approximate generator commitment without using binary variables. The generator output is linked to the relaxed commitment state by

$$P_{\min}^g s_t \leq P_t^g \leq P_{\max}^g s_t, \qquad \forall t \in T, \tag{11}$$

$$0 \leq s_t \leq 1, \qquad \forall t \in T. \tag{12}$$

To penalise switching of the relaxed generator state, an auxiliary variable is introduced:

$$d_t^s \geq s_t - s_{t-1}, \qquad \forall t \in T,\ t \geq 2, \tag{13}$$

$$d_t^s \geq -(s_t - s_{t-1}), \qquad \forall t \in T,\ t \geq 2, \tag{14}$$

$$d_t^s \geq 0, \qquad \forall t \in T,\ t \geq 2. \tag{15}$$

Similarly, let battery net power be defined as

$$P_t^b = P_t^d - P_t^c, \qquad \forall t \in T, \tag{16}$$

and use an auxiliary variable to represent its absolute value:

$$P_t^{b,\mathrm{abs}} \geq P_t^b, \qquad \forall t \in T, \tag{17}$$

$$P_t^{b,\mathrm{abs}} \geq -P_t^b, \qquad \forall t \in T, \tag{18}$$

$$P_t^{b,\mathrm{abs}} \geq 0, \qquad \forall t \in T. \tag{19}$$

The full regularised objective is then

$$\begin{aligned} \min \sum_{t \in T} \left(\pi_t P_t^{grid} + c_{\mathrm{fuel}} P_t^g + c_c P_t^c + c_d P_t^d \right) \Delta t + \gamma_{\mathrm{on}} \Delta t \sum_{t \in T} s_t \\ + \beta_{\mathrm{on}} \sum_{t \in T,\, t \geq 2} d_t^s + \gamma_{Pb} \Delta t \sum_{t \in T} P_t^{b,\mathrm{abs}} + \lambda \sum_{t \in T,\, t \geq 2} (P_t^g - P_{t-1}^g)^2. \end{aligned} \tag{20}$$

Together with (1)–(8), this yields a continuous optimisation problem that remains convex while encouraging sparse and less frequently switching generator behaviour.

A.6 Model C: Mixed-Integer Formulation

The mixed-integer model introduces binary variables to enforce exact operational logic.

Generator commitment

The generator output is linked to a binary commitment variable z_t:

$$P_{\min}^g z_t \leq P_t^g \leq P_{\max}^g z_t, \qquad \forall t \in T, \tag{21}$$

$$z_t \in \{0, 1\}, \qquad \forall t \in T. \tag{22}$$

Generator start-up logic

A binary start-up variable y_t is used to detect transitions from OFF to ON:

$$y_t \geq z_t - z_{t-1}, \qquad \forall t \in T,\ t \geq 2, \tag{23}$$

$$y_t \in \{0, 1\}, \qquad \forall t \in T,\ t \geq 2. \tag{24}$$

Battery operating modes

Binary mode variables are used to prevent simultaneous charging and discharging:

$$P_t^c \leq P_{\max}^b u_t^c, \qquad \forall t \in T, \tag{25}$$

$$P_t^d \leq P_{\max}^b u_t^d, \qquad \forall t \in T, \tag{26}$$

$$u_t^c + u_t^d \leq 1, \qquad \forall t \in T, \tag{27}$$

$$u_t^c, u_t^d \in \{0,1\}, \qquad \forall t \in T. \tag{28}$$

If a minimum battery operating power is imposed in the mixed-integer model, the following lower bounds may also be included:

$$P_t^c \geq P_{\min}^b u_t^c, \qquad \forall t \in T, \tag{29}$$

$$P_t^d \geq P_{\min}^b u_t^d, \qquad \forall t \in T. \tag{30}$$

The mixed-integer objective is

$$\begin{aligned} \min \sum_{t\in T} \left(\pi_t P_t^{grid} + c_{\text{fuel}} P_t^g + c_c P_t^c + c_d P_t^d\right) \Delta t \\ + c_{\text{on}} \Delta t \sum_{t\in T} z_t + c_{\text{start}} \sum_{t\in T,\, t\geq 2} y_t + \lambda \sum_{t\in T,\, t\geq 2} (P_t^g - P_{t-1}^g)^2. \end{aligned} \tag{31}$$

Together with (1)–(8), this formulation produces a mixed-integer linear or quadratic optimisation problem, depending on the value of λ.

A.7 Additional Performance Metrics

For completeness, the additional metrics used in Chapters 4–6 are listed below.

Generator usage

For the mixed-integer formulation, generator usage is measured by

$$\sum_{t\in T} z_t. \tag{32}$$

For the regularised convex formulation, the corresponding surrogate is

$$\sum_{t\in T} s_t. \tag{33}$$

Switching activity

In the mixed-integer formulation, switching activity is measured by

$$\sum_{t \in T,\, t \geq 2} y_t. \tag{34}$$

In the regularised convex formulation, the corresponding surrogate is

$$\sum_{t \in T,\, t \geq 2} d_t^s. \tag{35}$$

Distance to the mixed-integer solution

To quantify the similarity between the regularised and mixed-integer generator schedules, the distance metric is defined as

$$\mathrm{Dist}_g = \sum_{t \in T} \left| P^g_{t,\mathrm{reg}} - P^g_{t,\mathrm{mip}} \right|. \tag{36}$$

This metric is used in the parameter-sweep analysis to assess when the regularised convex formulation most closely approximates the mixed-integer solution.

B Additional Results

This appendix presents supplementary figures that support the discussion in Chapters 4–6. These plots are not essential to the main argument of the dissertation, but they provide additional evidence for the comparative behaviour of the convex, regularised convex, and mixed-integer formulations.

B.1 Supplementary synthetic benchmark result

The main synthetic benchmark results in Chapter 4 focus on generator dispatch, grid import, battery net power, and battery state of charge. The following figure provides an additional comparison of generator commitment behaviour.

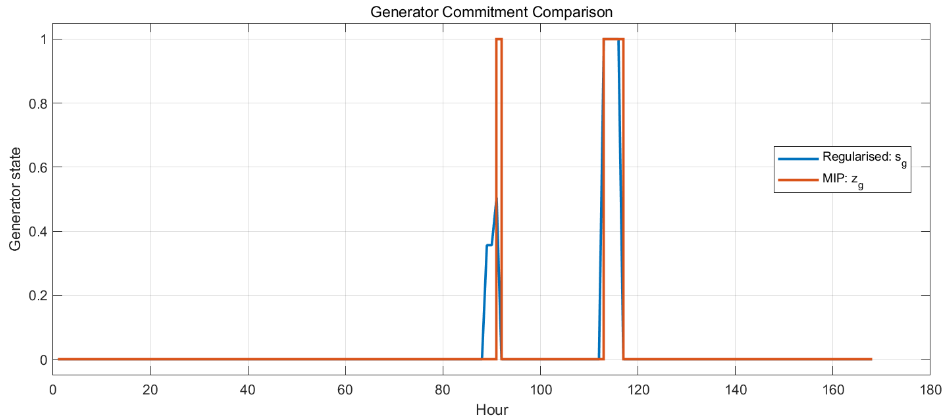

Figure 2: Supplementary generator commitment comparison for the synthetic benchmark.

The regularised convex model produces a continuous commitment variable, whereas the mixed-integer formulation enforces an exact binary on/off state. The overall trend shows that the regularised model captures the main activation pattern of the generator, but does not reproduce strict discrete commitment exactly.

B.2 Supplementary parameter-sweep time-series results

Chapter 5 presents the main trade-off curves for the parameter sweep. The figures in this subsection provide the corresponding time-series plots, which help illustrate how the generator schedule, grid import, battery power, and battery state of charge evolve as the regularisation parameters are varied.

Gamma sweep

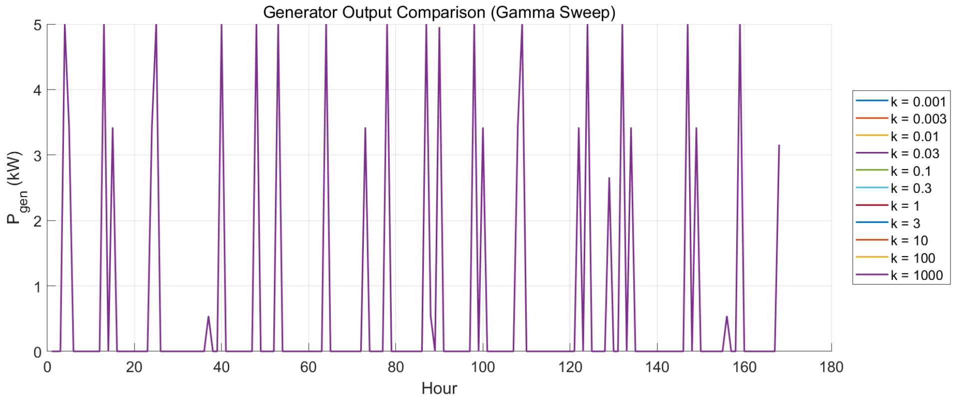

Figure 3: Supplementary generator-output trajectories for the γ sweep.

Most trajectories overlap substantially, showing that the main generator activation periods are broadly preserved as γ varies. This indicates that, within the tested parameter range, the on-time penalty affects generator usage mainly at the margin rather than fundamentally changing the overall dispatch structure.

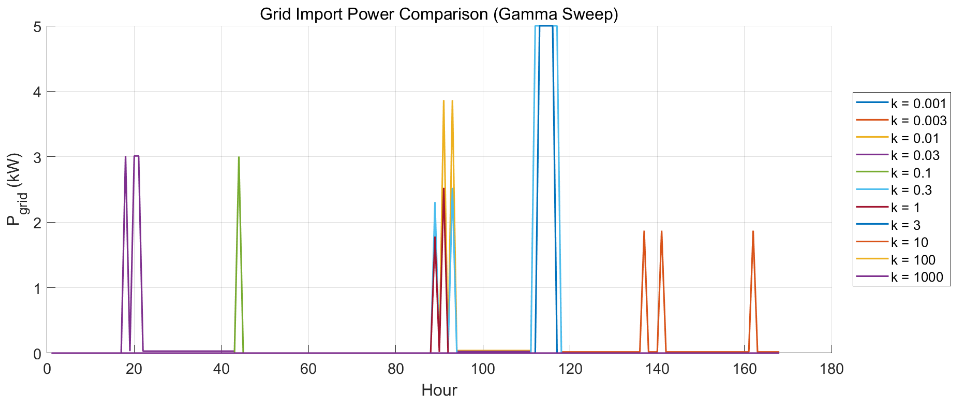

Figure 4: Supplementary grid-import trajectories for the .

Grid import remains zero for most of the 168-hour horizon, with only a limited number of distinct import events. As γ varies, the main differences appear in the timing and magnitude of these peaks rather than in a uniform increase in grid dependence. This suggests that the generator-usage penalty mainly redistributes when grid support is required, instead of fundamentally changing the overall sparse dispatch structure.

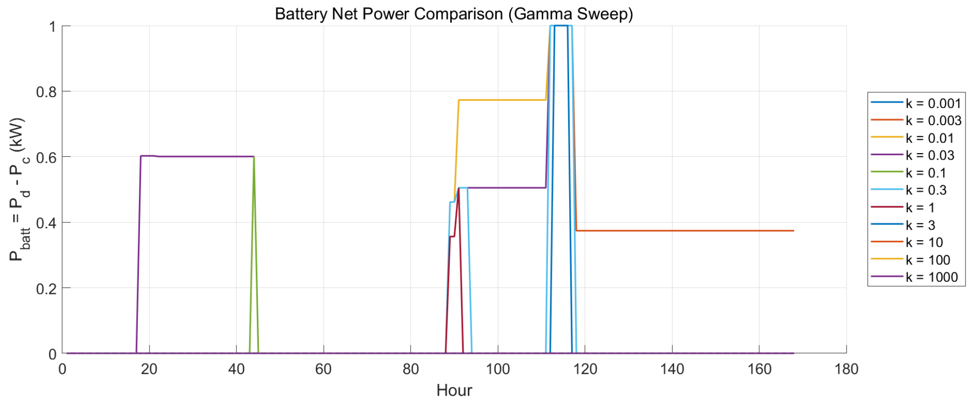

Figure 5: Supplementary battery net-power trajectories for the γ sweep.

Most trajectories remain at zero for large parts of the 168-hour horizon, with battery activity concentrated in only a few discharge intervals. As γ varies, the main differences appear in the timing, duration, and magnitude of these discharge events, indicating that the generator-usage penalty mainly redistributes when battery support is used rather than producing a uniform increase in battery utilisation.

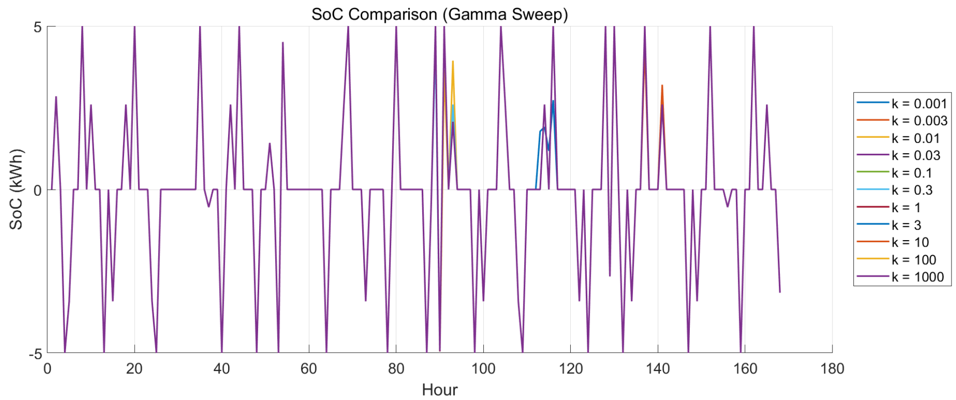

Figure 6: Supplementary battery state-of-charge trajectories for the γ sweep.

The trajectories are almost completely overlapping across the tested values of γ, with only minor local deviations in a few short intervals. This indicates that the overall battery energy trajectory is highly robust to changes in the generator-usage penalty, and that the influence of γ is limited mainly to local dispatch details rather than the global SoC evolution.

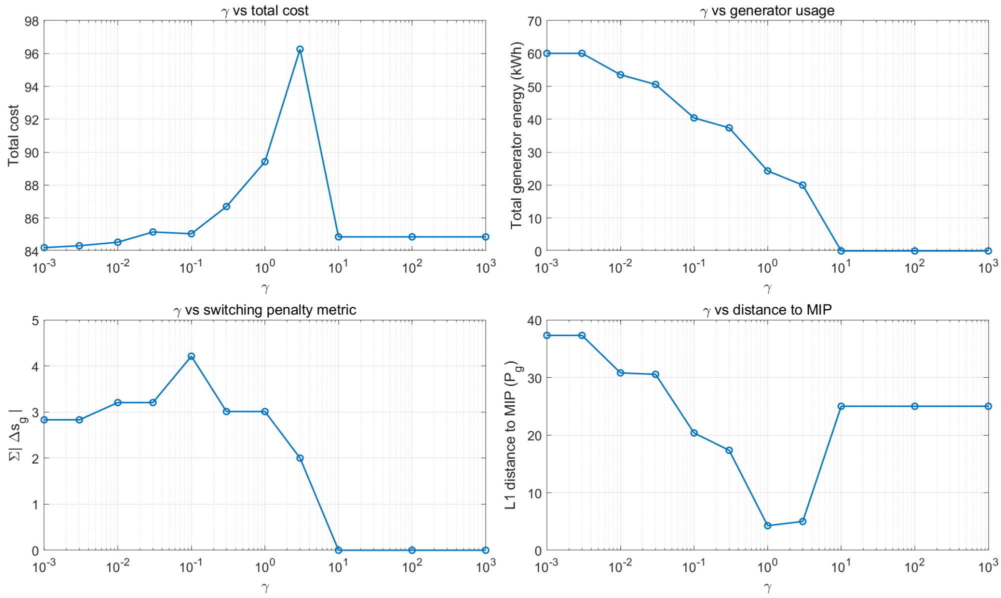

Figure 7: Supplementary summary metrics for the γ sweep.

The aggregate effects of γ are more clearly visible here than in the individual time-series trajectories. As γ increases, the total cost decreases markedly over an intermediate parameter range and then reaches a plateau, while the switching-related metric falls sharply and approaches zero for sufficiently large γ. By contrast, total generator usage and the L1 distance to the mixed-integer solution change only slightly, indicating that the main role of the generator-usage penalty is to regularise the scheduling structure and suppress switching-related behaviour rather than to substantially alter overall generator utilisation.

Beta sweep

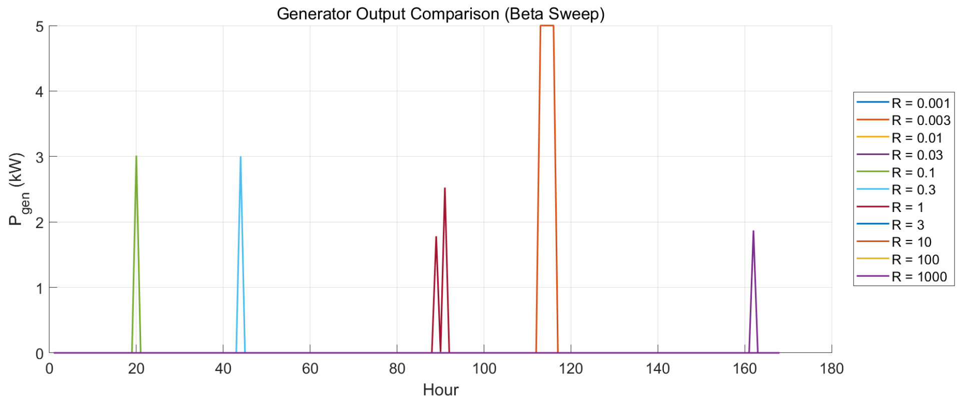

Figure 8: Supplementary generator-output trajectories for the β sweep.

The trajectories differ substantially across the tested values of β, with generation peaks appearing at different times and with different magnitudes. This indicates that the switching penalty strongly influences the temporal placement of generator operation. Although generator dispatch remains sparse in all cases, the timing of active intervals is clearly reshaped as β varies.

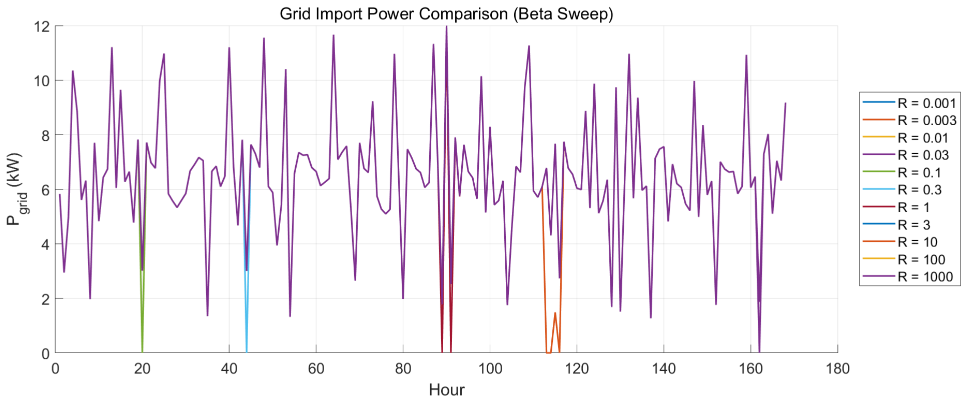

Figure 9: Supplementary grid-import trajectories for the β sweep.

The trajectories show a strong dependence on β. For several parameter values, grid import occurs only in a few isolated intervals, whereas at larger β one trajectory becomes dominant over almost the entire 168-hour horizon, with sustained and relatively high grid import. This indicates that a sufficiently strong switching penalty can shift the system into a different scheduling regime, where reduced generator switching flexibility is compensated by much more persistent reliance on the grid.

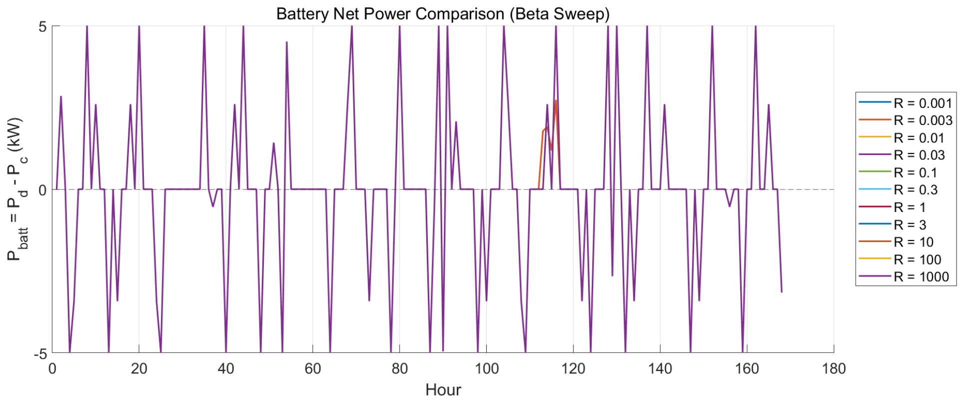

Figure 10: Supplementary battery net-power trajectories for the β sweep.

The trajectories are highly overlapping across the tested values of β, indicating that the overall charge/discharge pattern is largely preserved. Only small local deviations appear in a few short intervals, suggesting that the switching penalty affects battery dispatch mainly at the margin rather than fundamentally changing its overall temporal structure.

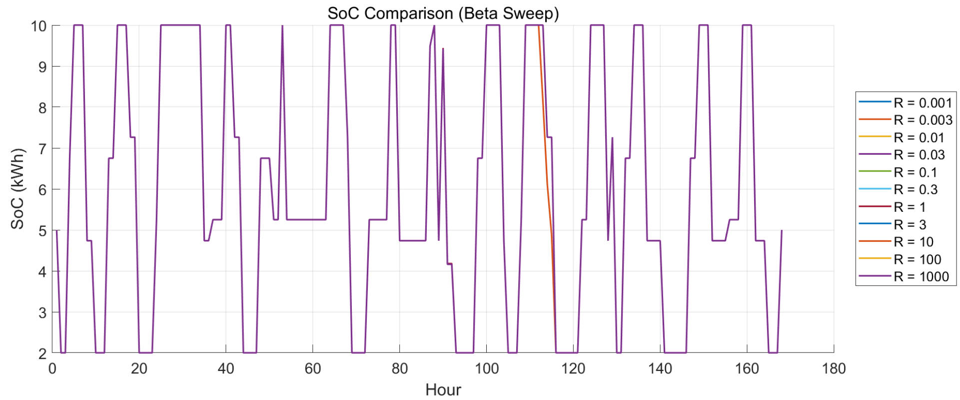

Figure 11: Supplementary battery state-of-charge trajectories for the β sweep.

The curves are almost completely overlapping for all tested values of β, with only negligible local deviations. This indicates that the cumulative battery energy trajectory is highly robust to variations in the switching penalty, even though β may alter dispatch decisions elsewhere in the system.

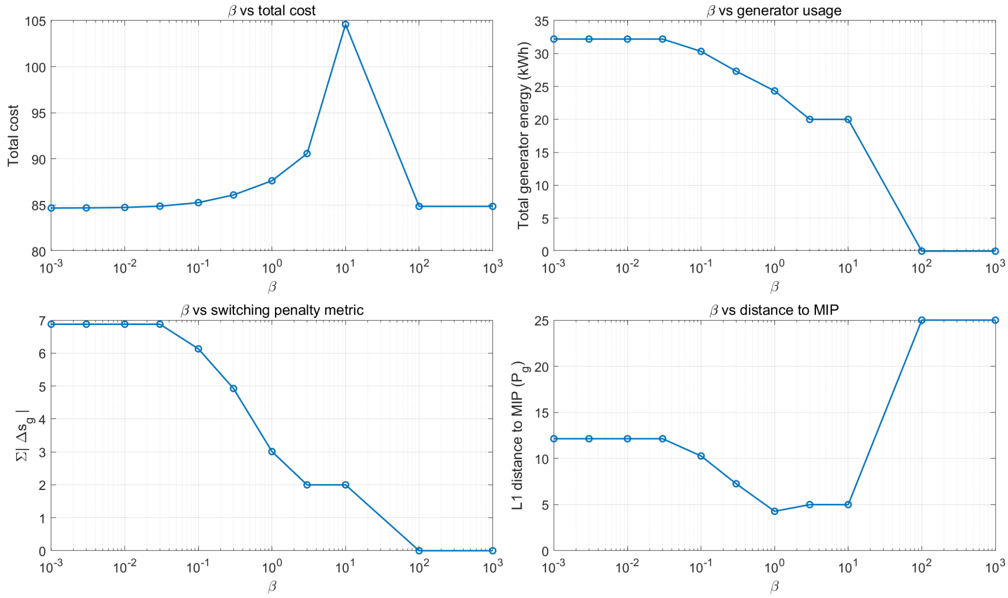

Figure 12: Supplementary summary metrics for the β sweep.

As β increases, both total generator usage and the switching-related metric decrease substantially, confirming that the switching penalty effectively suppresses generator activity and switching behaviour. However, the total cost varies non-monotonically, increasing towards an intermediate peak before falling again at larger β. The L1 distance to the mixed-integer solution also decreases at first and reaches a minimum at an intermediate value of β, but rises sharply again when β becomes too large. This indicates that moderate values of β provide the best compromise between reduced switching, acceptable cost, and similarity to the mixed-integer benchmark.

B.3 Supplementary UK-data input plots

Chapter 6 uses real UK market price data while retaining the same load profile as in the synthetic benchmark. The following input-data figures are included here for completeness.

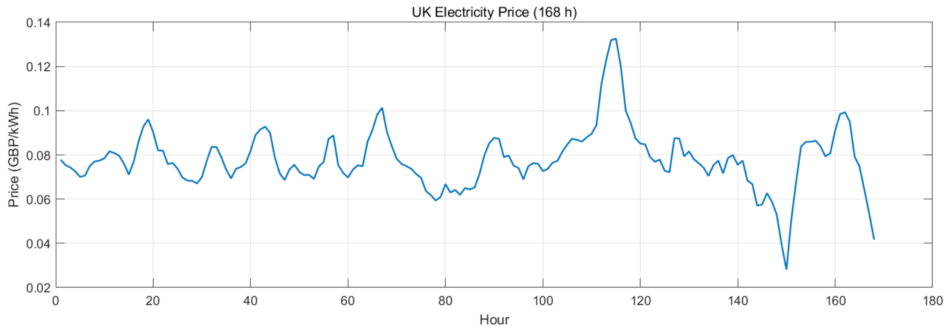

Figure 13: Hourly UK electricity price series over the selected 168-hour validation period. The figure shows the realistic time variation in wholesale market prices used as the grid electricity cost in Chapter 6.

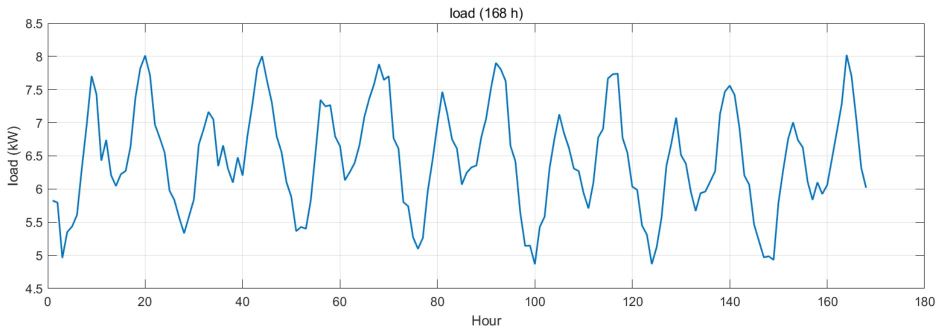

Figure 14: Load profile used in the 168-hour experiments. The same demand trajectory is retained in the UK-data validation study so that the impact of replacing the synthetic price signal can be isolated more clearly.

C Code Structure

This appendix summarises the organisation of the code developed for this project. The implementation is divided into three main parts: experiment folders, optimisation models, and post-processing scripts. This structure separates model execution, result storage, and figure generation, which improves clarity, reproducibility, and maintainability.

C.1 Overall Folder Structure

```
experiments/
  exp_01_synthetic_baseline/
```

```
    figures/
    outputs/
  exp_02_parameter_sweep/
    figures/
    outputs/
  exp_03_uk_validation/
    figures/
    outputs/
    uk_grid_data/

models/
  convex/
    model.mod
    run_exp01.run
    run_exp03.run
  mip/
    model.mod
    run_exp01.run
    run_exp03.run
  regularised/
    model.mod
    run_exp01.run
    run_exp02_sweep.run
    run_exp03.run
  without_optimization/
    model.mod
    run.run

postprocessing/
  compare_models.m
  make_dat.m
  plot_regularised_sweep_pg.m
  tradeoff_plots.m
  uk_analysis.m
```

C.2 Description of the Code Structure

The codebase is organised into three main directories: experiments, models, and postprocessing. This separation allows the workflow to be divided into experiment execution, optimisation model definition, and result analysis.

Experiments The `experiments` directory stores the outputs and generated figures for the numerical studies carried out in this project. It is divided into three subdirectories:

- `exp_01_synthetic_baseline`: synthetic baseline experiment;
- `exp_02_parameter_sweep`: regularisation parameter sweep study;
- `exp_03_uk_validation`: UK-data validation study.

Each experiment folder contains the relevant numerical outputs, and where applicable, the figures produced from those outputs. In the UK validation study, the input market data are additionally stored in a dedicated subfolder.

Models The `models` directory contains the AMPL implementations of the optimisation formulations used in the dissertation. It includes four model categories:

- `convex`: baseline convex optimisation model;
- `mip`: mixed-integer programming formulation;
- `regularised`: regularised convex formulation used to approximate switching behaviour;
- `without_optimization`:reference case without optimisation-based scheduling, used as a baseline to calculate the total cost when the load is supplied entirely by the grid.

Each model folder contains the corresponding `model.mod` file together with the required run files for the relevant experiments.

Post-processing The `postprocessing` directory contains the MATLAB scripts used to prepare input data, compare optimisation results, and generate the figures presented in the dissertation. The role of each script is summarised below.

MATLAB script	Function
make_dat.m	Generates the input data files used in the experiments, including the synthetic demand, price, and system parameter settings required by the AMPL models.
compare_models.m	Reads and compares the outputs of different optimisation formulations, such as the convex, MIP, and regularised models, in order to evaluate differences in cost, dispatch behaviour, and operating profiles.
plot_..._sweep_pg.m	Produces figures for the generator-output behaviour under the regularisation parameter sweep, showing how the generator dispatch changes as the penalty parameters are varied.
tradeoff_plots.m	Generates the trade-off analysis figures used to compare solution quality and approximation behaviour across different regularisation settings, helping to visualise the balance between cost and switching-related penalties.
uk_analysis.m	Processes and visualises the results of the UK-data validation case, including the analysis of dispatch behaviour and cost trends under real electricity price data.

C.3 Code Repository

The source code developed for this project is available in the following public GitHub repository:

`https://github.com/ZXURUI-pickle/ELE323-individual-project`

The repository contains the optimisation models, experiment files, and MATLAB scripts used in this dissertation. In particular, it includes the convex, regularised convex, and mixed-integer formulations, together with the post-processing scripts used to generate the reported figures and comparative results.

The code has been uploaded to support reproducibility and to provide access to the implementation details underlying the work presented in this dissertation.